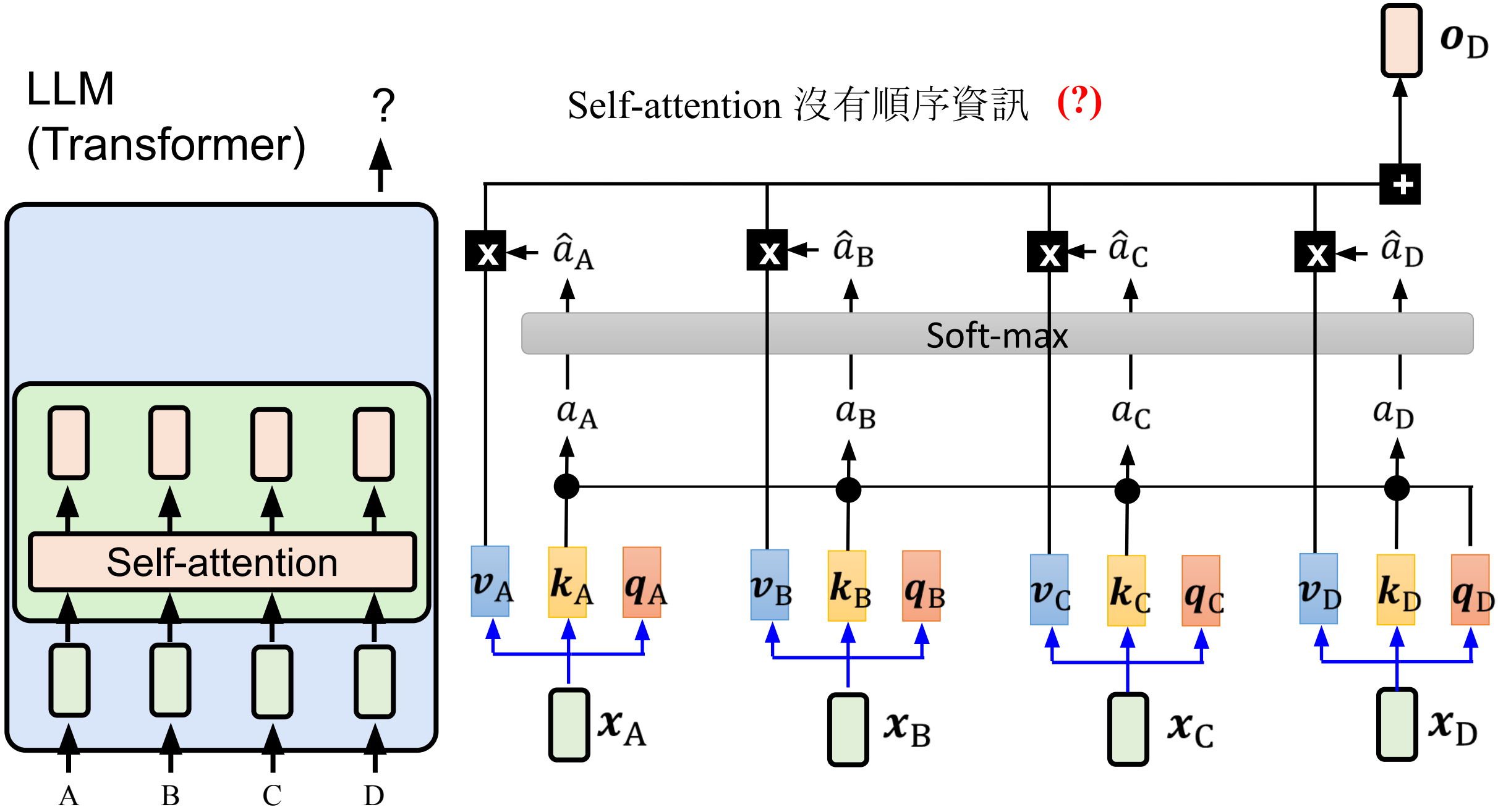
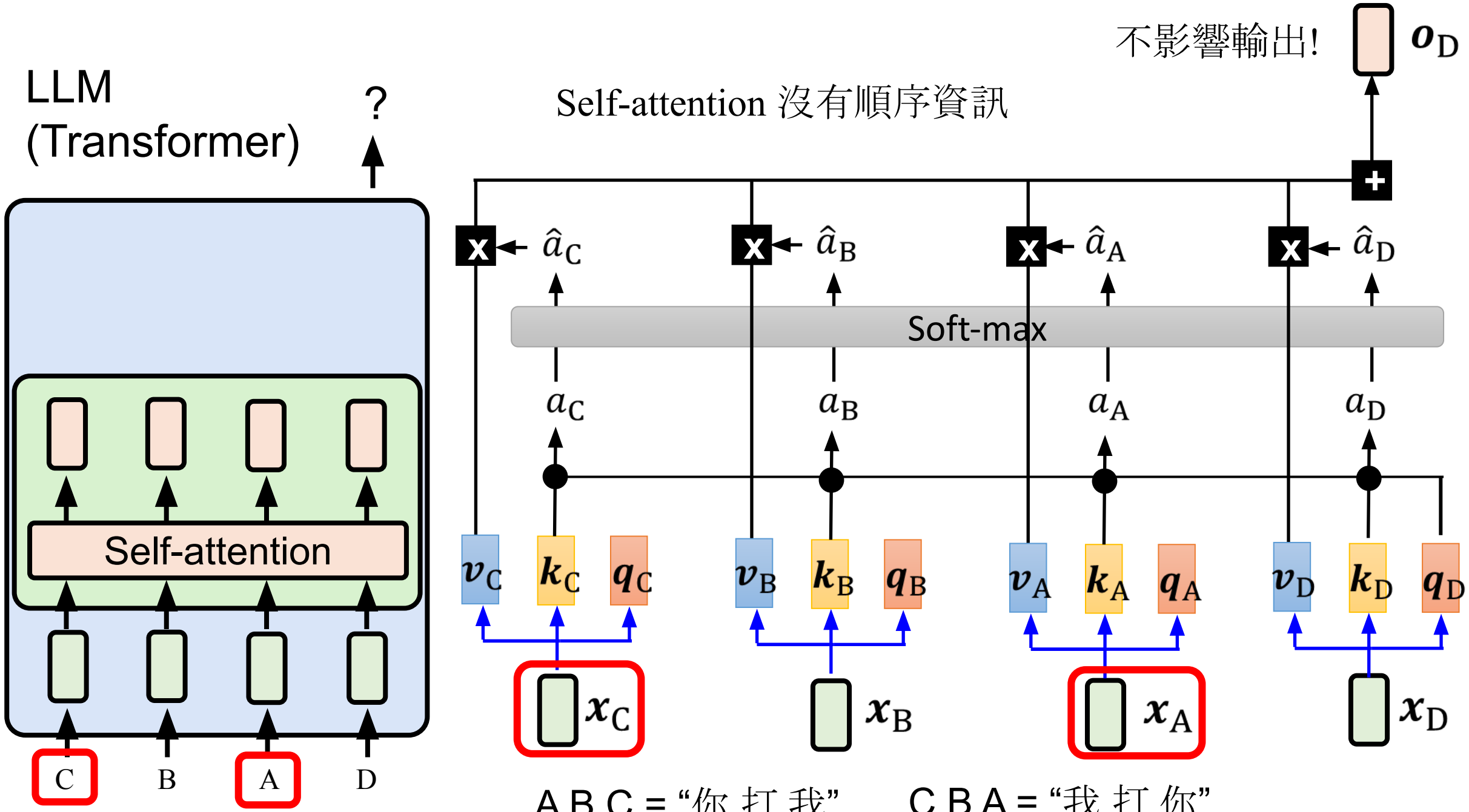
Positional Embedding

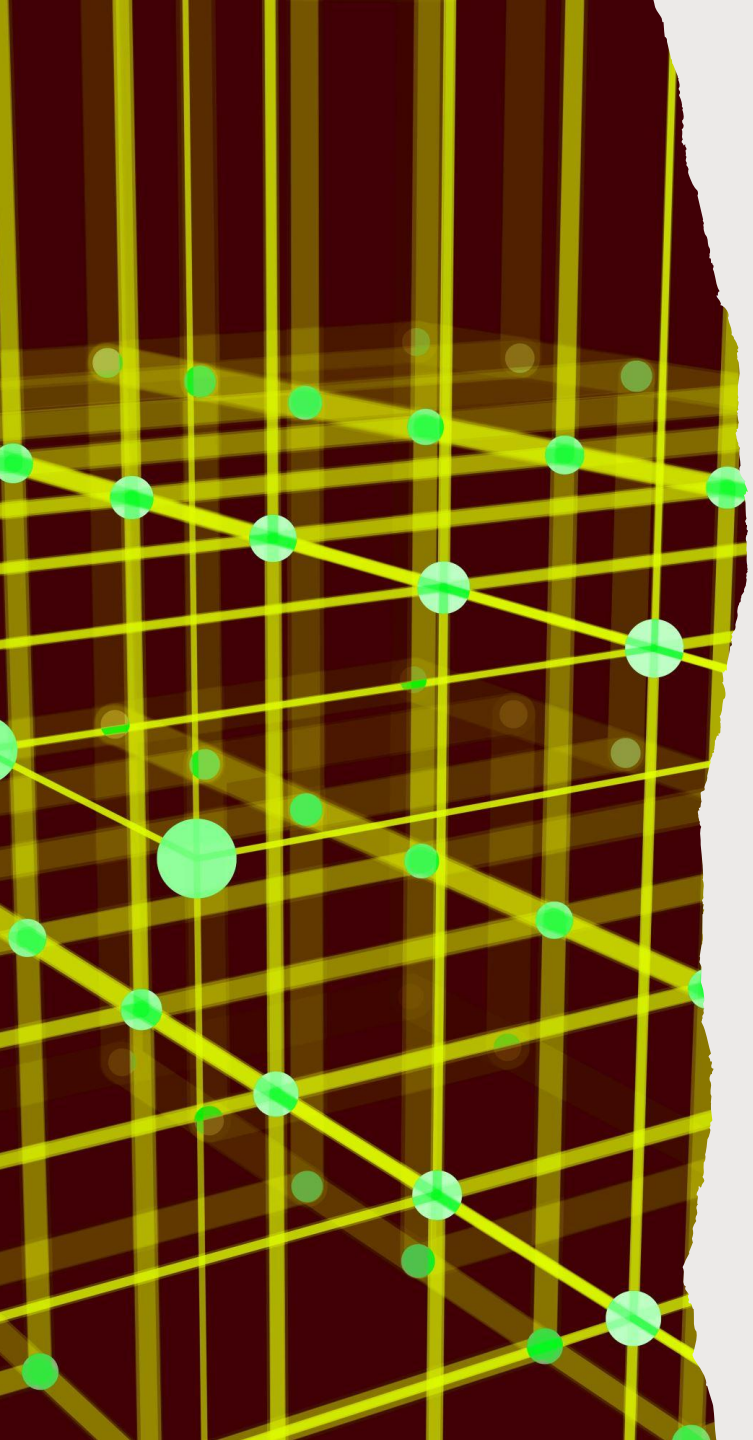
如何讓 Transformer 知道輸入 Token 的順序？


LLM
(Transformer)
?
Self-attention
A
B
C
D
Self-attention 沒有順序資訊 (?)
o_D
$\hat{a}_A$
$\hat{a}_B$
$\hat{a}_C$
$\hat{a}_D$
Soft-max
a_A
a_B
a_C
a_D
v_A k_A q_A
v_B k_B q_B
v_C k_C q_C
v_D k_D q_D
x_A
x_B
x_C
x_D


LLM
(Transformer)
?
Self-attention
C
B
A
D
Self-attention 沒有順序資訊
不影響輸出!
o_D
$\hat{a}_C$
$\hat{a}_B$
$\hat{a}_A$
$\hat{a}_D$
Soft-max
a_C
a_B
a_A
a_D
v_C k_C q_C
v_B k_B q_B
v_A k_A q_A
v_D k_D q_D
x_C
x_B
x_A
x_D
A B C = “你 打 我”
C B A = “我 打 你”

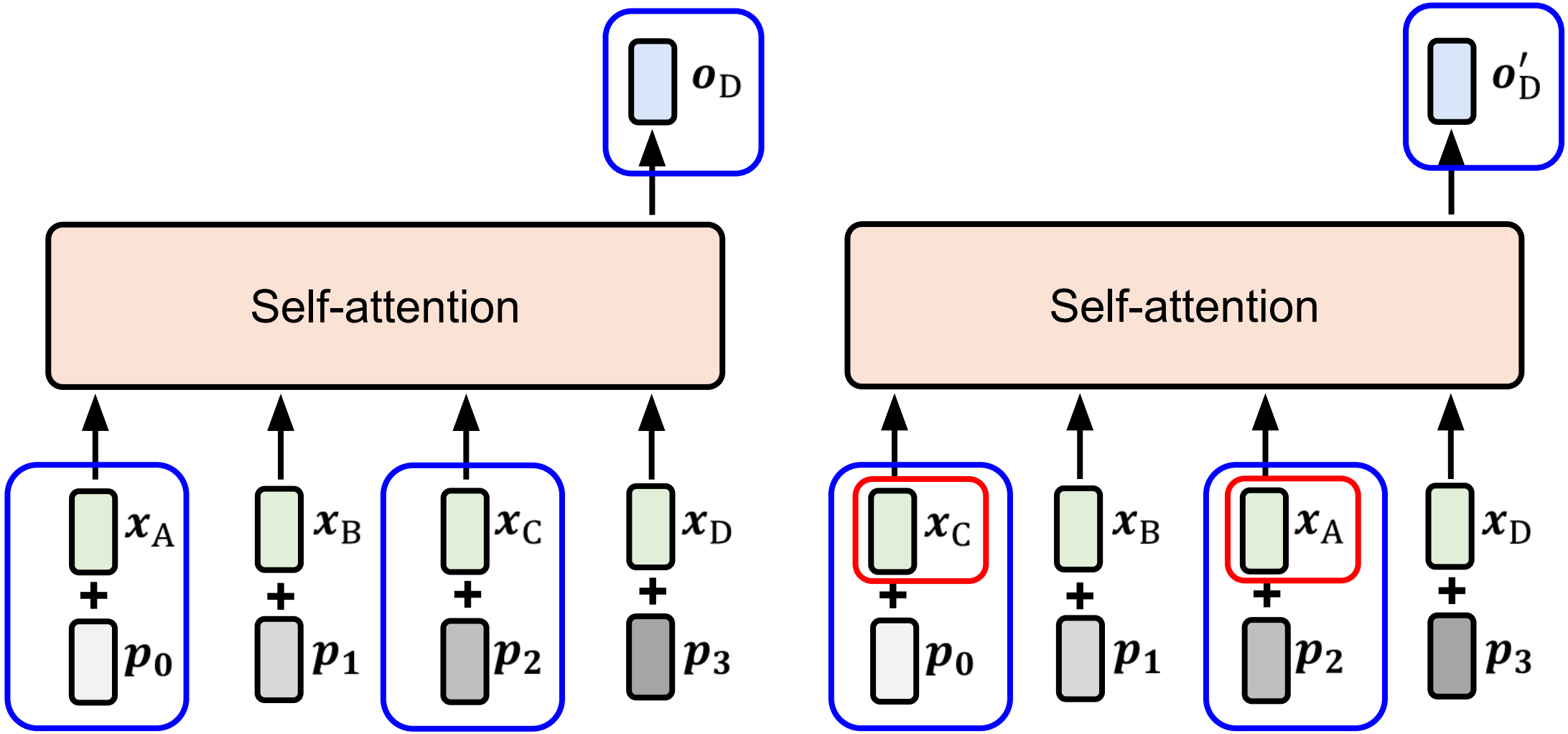
Outline

- Absolute Positional Embedding
- Relative Positional Embedding
- RoPE
- Train short, test long
- No Positional Embedding?!

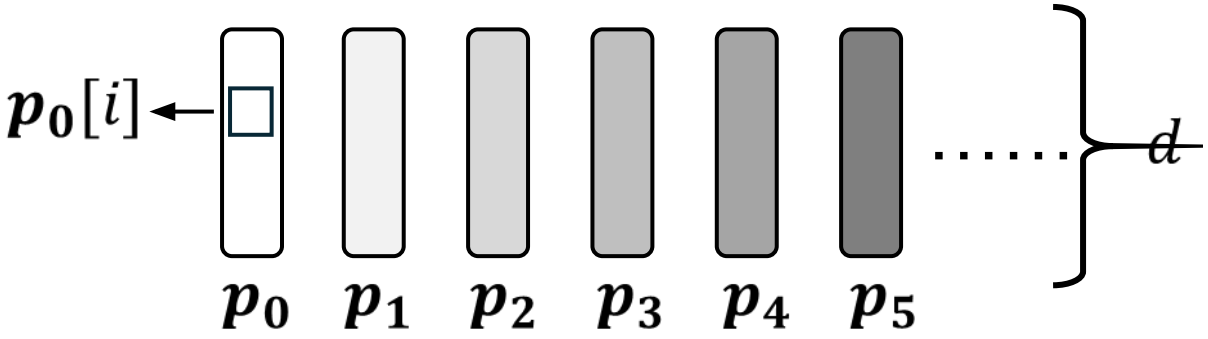


Absolute Positional Embedding

Absolute Positional Embedding



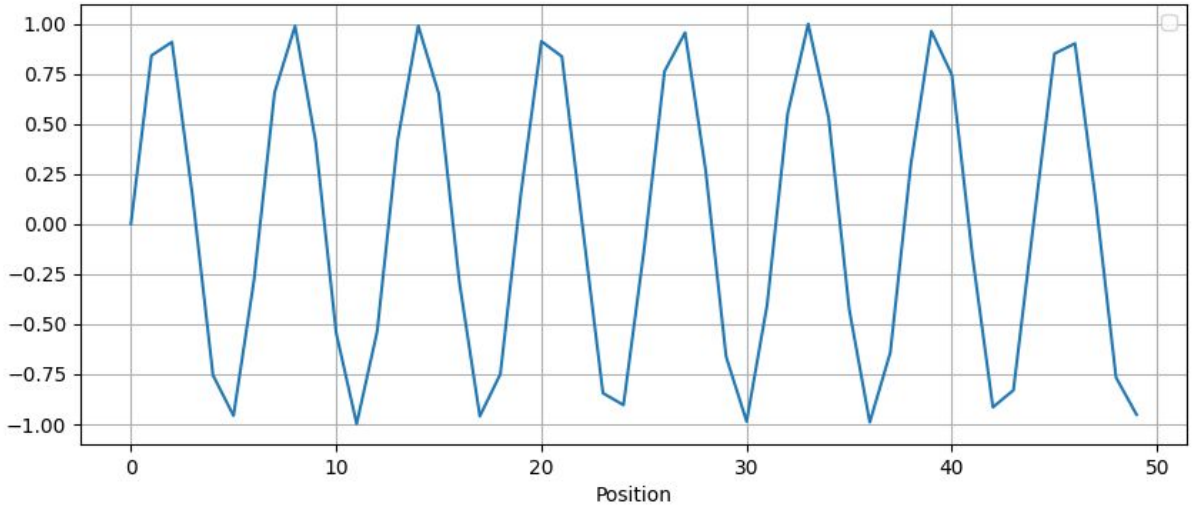
Sinusoidal Positional Embedding



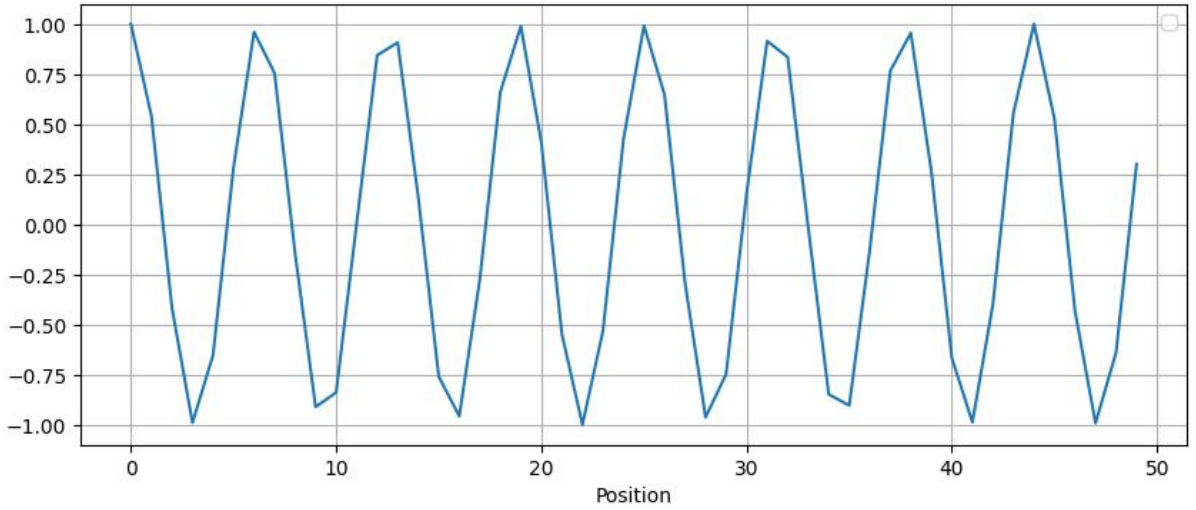
$$p_k[2i] = sin\left(\frac{k}{10000^{2i/d}}\right)$$

$$p_k[2i+1] = cos\left(\frac{k}{10000^{2i/d}}\right)$$

$p_0[0] \dots p_{49}[0]$

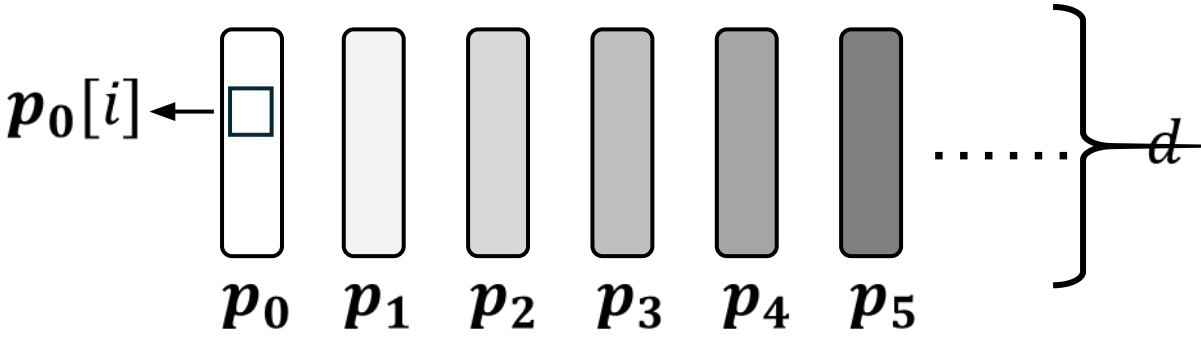


$p_0[1] \dots p_{49}[1]$



https://arxiv.org/abs/1706.03762

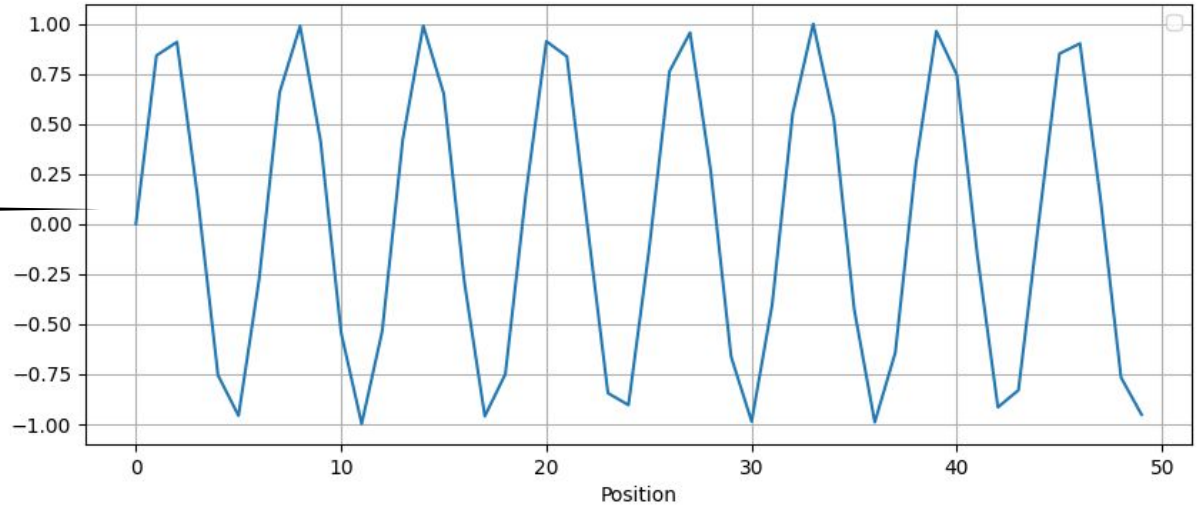
Sinusoidal Positional Embedding



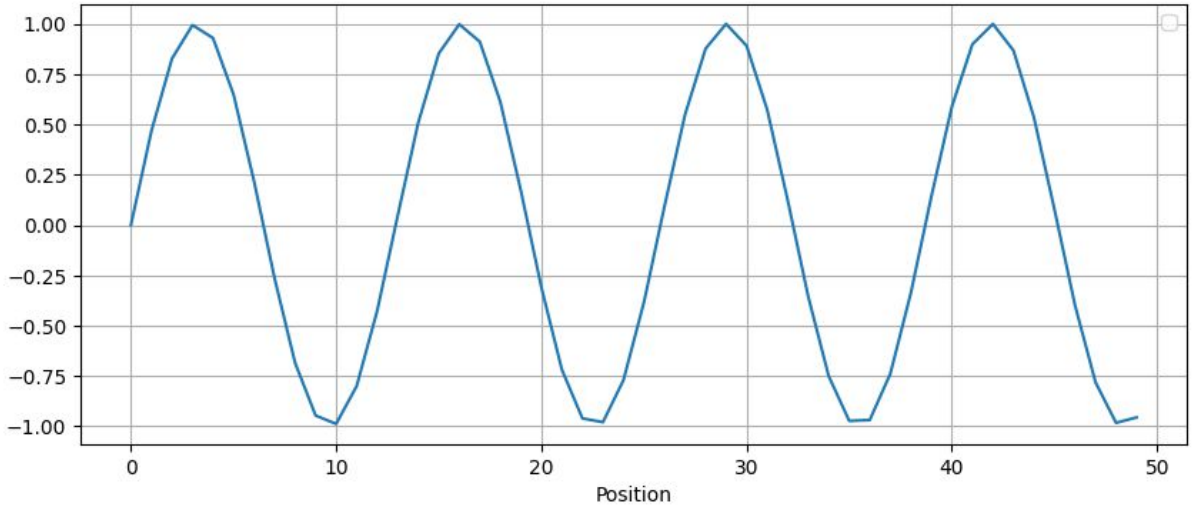
$$p_k[2i] = sin\left(\frac{k}{10000^{2i/d}}\right)$$

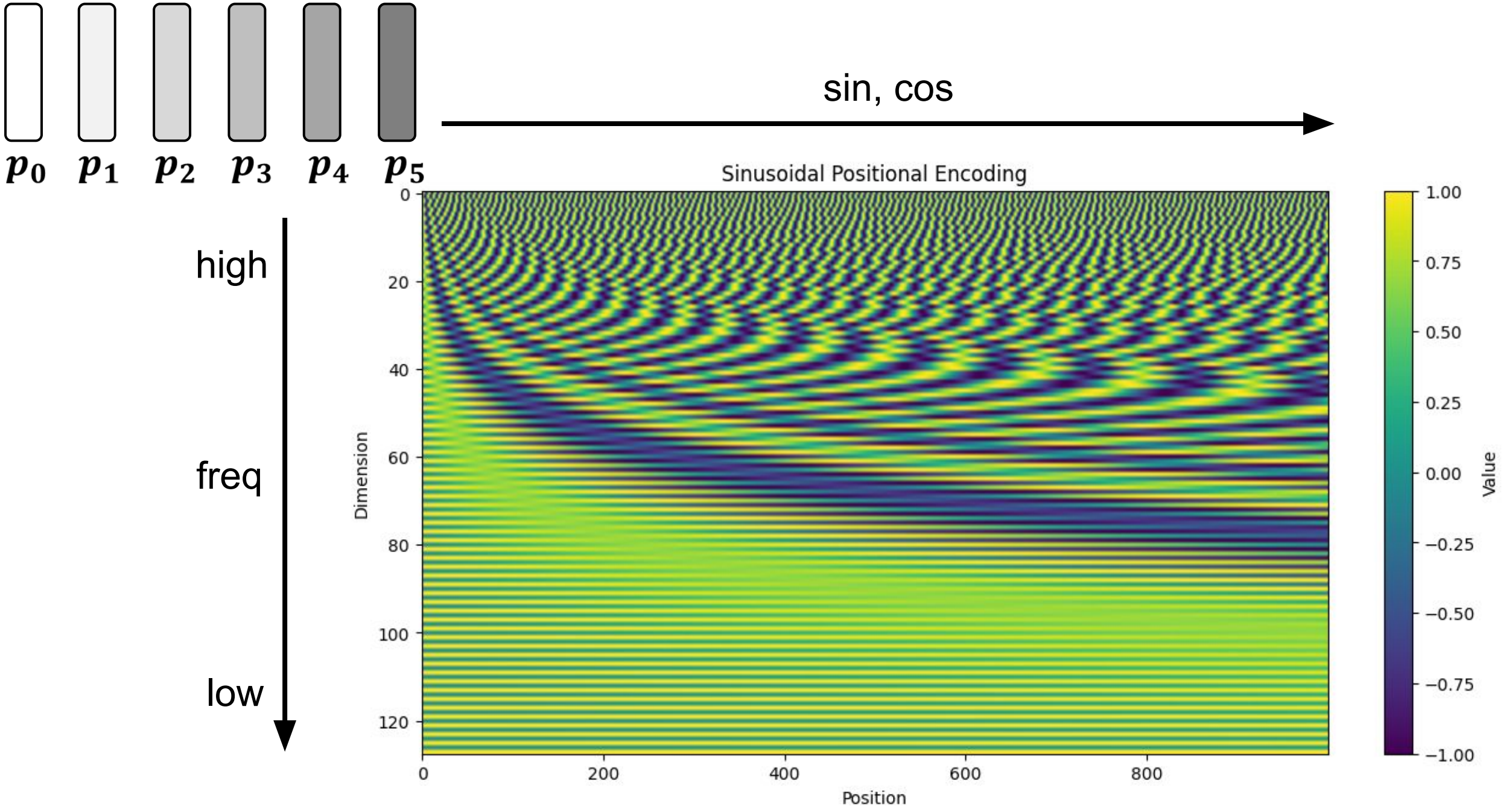
$$p_k[2i+1] = cos\left(\frac{k}{10000^{2i/d}}\right)$$

$p_0[0] \dots p_{49}[0]$



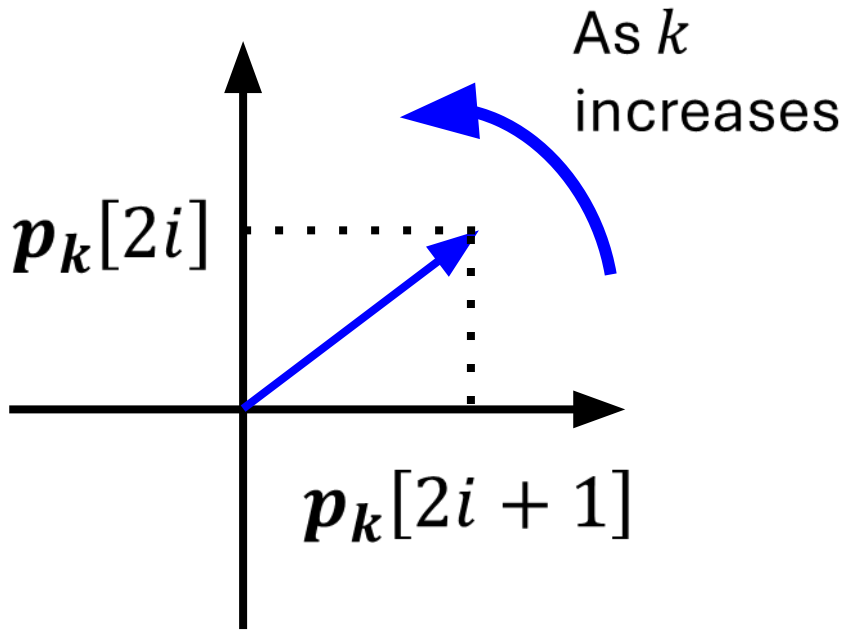
$p_0[10] \dots p_{49}[10]$




p_0 p_1 p_2 p_3 p_4 p_5
sin, cos
high
freq
low
Sinusoidal Positional Encoding
Dimension
Position
Value
0
20
40
60
80
100
120
0
200
400
600
800
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

$$\boldsymbol{p_k}[2i] = sin\left(\frac{k}{10000^{2i/d}}\right)$$

$$\boldsymbol{p_k}[2i+1] = cos\left(\frac{k}{10000^{2i/d}}\right)$$



$$\frac{k}{10000^{2i/d}} = 2\pi$$

$$k = 2\pi \cdot 10000^{2i/d}$$

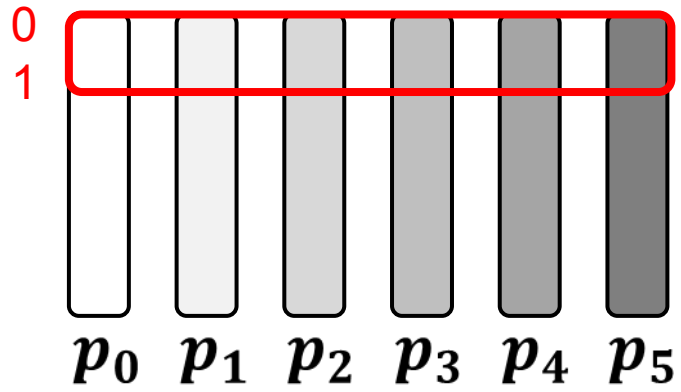
$$d = 128$$

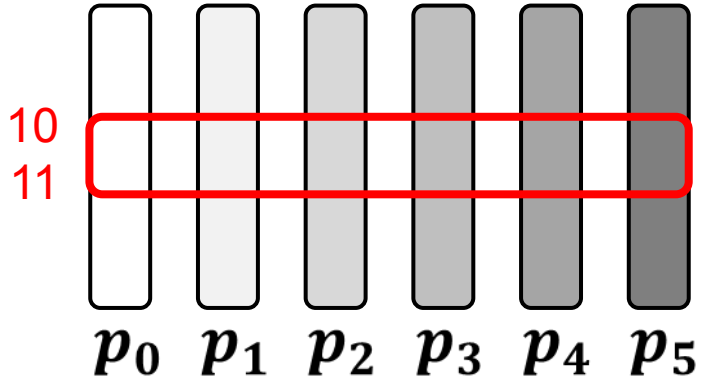
$$i = 0, \ldots, \frac{d}{2} - 1 = 0, \ldots, 63$$

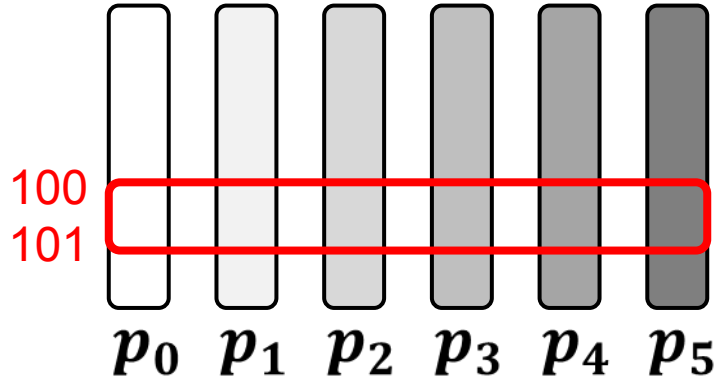
$i = 0$ (0,1 dim) `6.3`

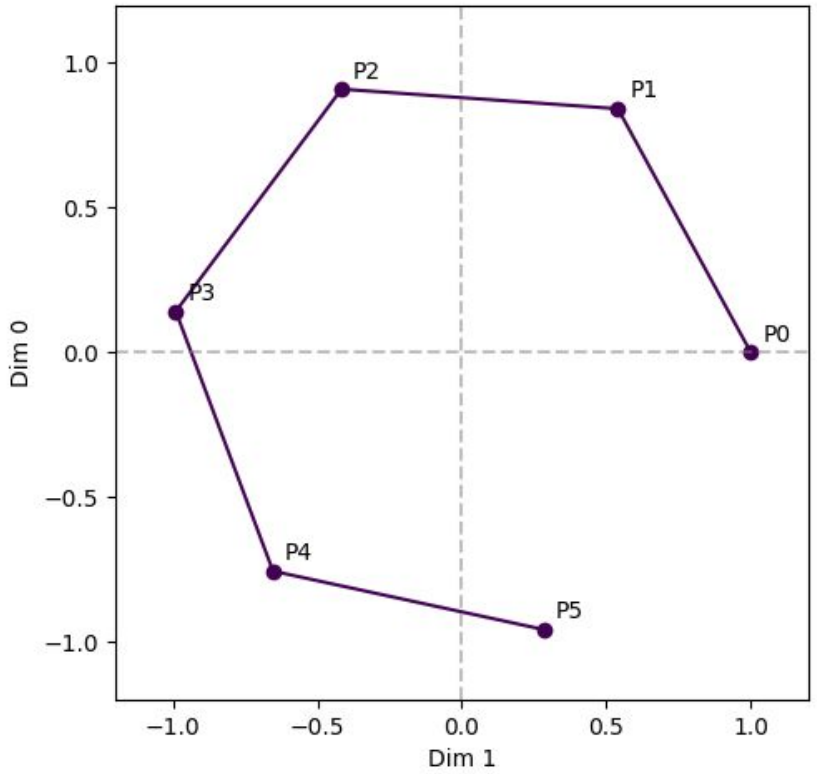
$i = 32$ (64,65 dim) `628.3`

$i = 63$ (126,127 dim) `54410.1`

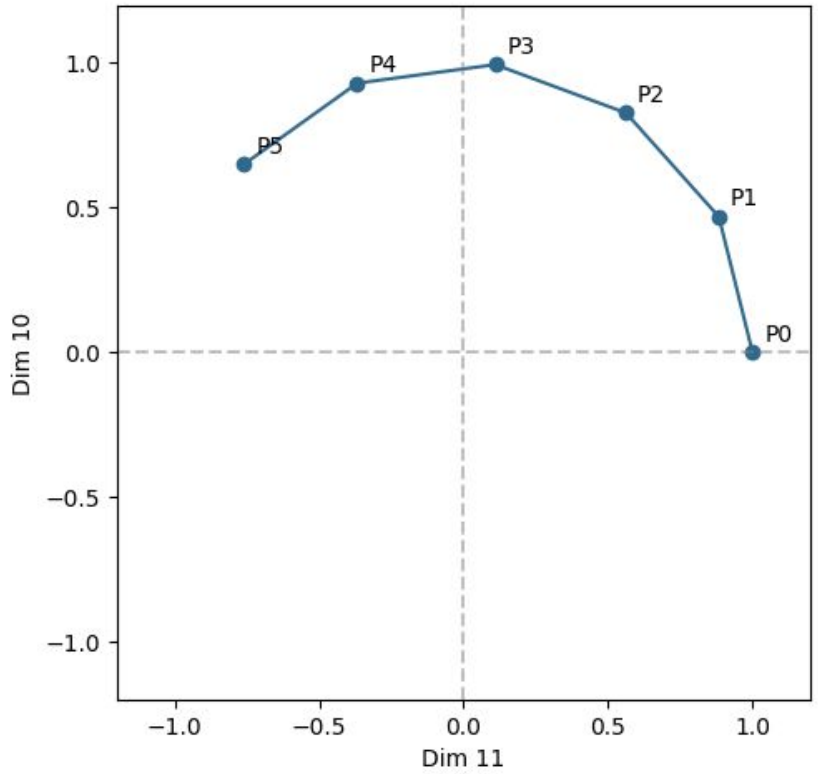

0
1
p_0 p_1 p_2 p_3 p_4 p_5


10
11
p_0 p_1 p_2 p_3 p_4 p_5

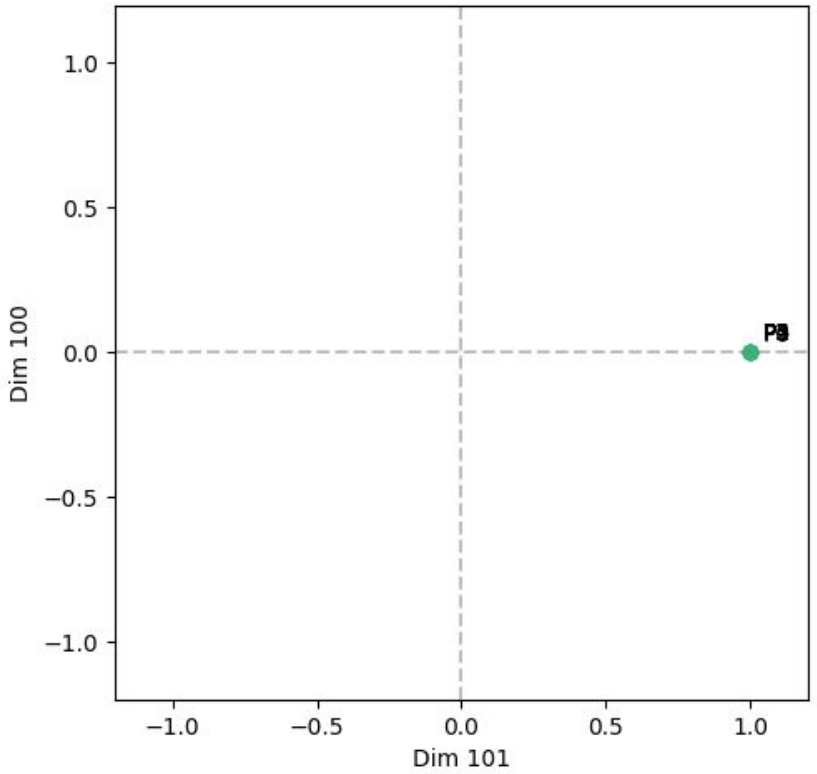

100
101
p_0 p_1 p_2 p_3 p_4 p_5


P2
P1
P3
P0
P4
P5
Dim 0
Dim 1
1.0
0.5
0.0
−0.5
−1.0

秒針


P4
P3
P2
P5
P1
P0
Dim 10
Dim 11
1.0
0.5
0.0
−0.5
−1.0

分針


Dim 100
Dim 101
1.0
0.5
0.0
−0.5
−1.0

時針

3.5 Positional Encoding

Since our model contains no recurrence and no convolution, in order for the model to make use of the order of the sequence, we must inject some information about the relative or absolute position of the tokens in the sequence. To this end, we add "positional encodings" to the input embeddings at the bottoms of the encoder and decoder stacks. The positional encodings have the same dimension d_{model} as the embeddings, so that the two can be summed. There are many choices of positional encodings, learned and fixed [9].

In this work, we use sine and cosine functions of different frequencies:

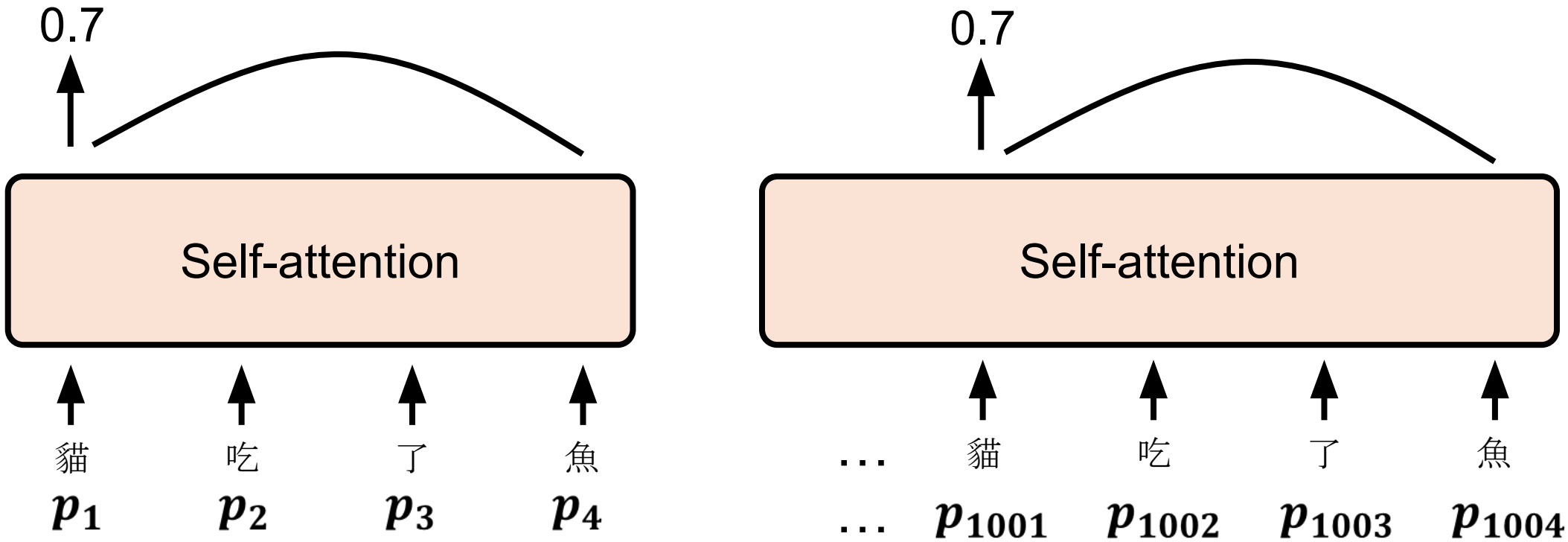
$$PE_{(pos,2i)} = sin(pos/10000^{2i/d_{\text{model}}})$$
$$PE_{(pos,2i+1)} = cos(pos/10000^{2i/d_{\text{model}}})$$

where pos is the position and i is the dimension. That is, each dimension of the positional encoding corresponds to a sinusoid. The wavelengths form a geometric progression from 2π to $10000 \cdot 2\pi$. We chose this function because we hypothesized it would allow the model to easily learn to attend by relative positions, since for any fixed offset k, PE_{pos+k} can be represented as a linear function of PE_{pos}.

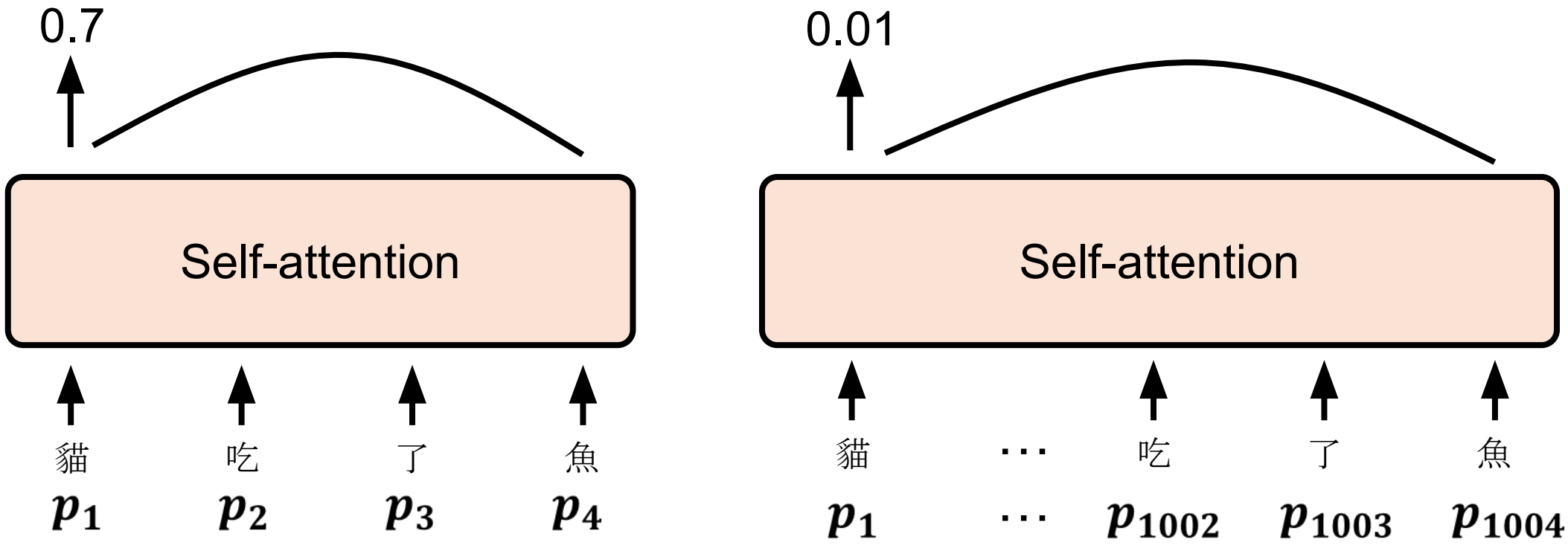
We also experimented with using learned positional embeddings [9] instead, and found that the two versions produced nearly identical results (see Table 3 row (E)). We chose the sinusoidal version because it may allow the model to extrapolate to sequence lengths longer than the ones encountered during training.

https://arxiv.org/abs/1706.03762

Relative positioning is crucial!



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https://arxiv.org/abs/1706.03762

Sinusoidal Positional Embedding

$$\boldsymbol{p_{k+r}} = M_r \boldsymbol{p_k}$$

$$\boldsymbol{p_4} = M_3 \boldsymbol{p_1}$$

$$\boldsymbol{p_{14}} = M_3 \boldsymbol{p_{11}}$$

$$\boldsymbol{p_{104}} = M_3 \boldsymbol{p_{101}}$$

$$\boldsymbol{p_k}[2i] = sin\left(\frac{k}{10000^{2i/d}}\right)_z \qquad \boldsymbol{p_k}[2i+1] = cos\left(\frac{k}{10000^{2i/d}}\right)_z$$

$$\boldsymbol{p_{k+r}}[2i] = sin\left(\frac{k+r}{z}\right) \qquad \boldsymbol{p_k}[2i] = sin\left(\frac{k}{z}\right)$$

$$\boldsymbol{p_{k+r}}[2i+1] = cos\left(\frac{k+r}{z}\right) \qquad \boldsymbol{p_k}[2i+1] = cos\left(\frac{k}{z}\right)$$

$$sin(a+b) = sin(a)cos(b) + cos(a)sin(b)$$

$$cos(a+b) = cos(a)cos(b) - sin(a)sin(b)$$

Back to Sinusoidal Positional Embedding

$$\boldsymbol{p}_{k+r} = M_r \boldsymbol{p}_k$$

$$\boldsymbol{p}_k[2i] = sin\left(\frac{k}{10000^{2i/d}}\right) \qquad \boldsymbol{p}_k[2i+1] = cos\left(\frac{k}{10000^{2i/d}}\right)$$

$$z \qquad\qquad z$$

$$\boldsymbol{p}_{k+r}[2i] = sin\left(\frac{k+r}{z}\right) = sin\left(\frac{k}{z}\right)cos\left(\frac{r}{z}\right) + cos\left(\frac{k}{z}\right)sin\left(\frac{r}{z}\right)$$

$$\boldsymbol{p}_{k+r}[2i+1] = cos\left(\frac{k+r}{z}\right) = cos\left(\frac{k}{z}\right)cos\left(\frac{r}{z}\right) - sin\left(\frac{k}{z}\right)sin\left(\frac{r}{z}\right)$$

$$\boldsymbol{p}_k[2i] = sin\left(\frac{k}{z}\right)$$

$$\boldsymbol{p}_k[2i+1] = cos\left(\frac{k}{z}\right)$$

$$sin(a+b) = sin(a)cos(b) + cos(a)sin(b)$$

$$cos(a+b) = cos(a)cos(b) - sin(a)sin(b)$$

$$\boldsymbol{p_{k+r}} = M_r \boldsymbol{p_k} \qquad \boldsymbol{p_k}[2i] = sin\left(\frac{k}{10000^{2i/d}}\right)_z \qquad \boldsymbol{p_k}[2i+1] = cos\left(\frac{k}{10000^{2i/d}}\right)_z$$

$$\boldsymbol{p_{k+r}}[2i] = sin\left(\frac{k+r}{z}\right) = sin\left(\frac{k}{z}\right)cos\left(\frac{r}{z}\right) + cos\left(\frac{k}{z}\right)sin\left(\frac{r}{z}\right)$$

$$= \boldsymbol{p_k}[2i]cos\left(\frac{r}{z}\right) + \boldsymbol{p_k}[2i+1]sin\left(\frac{r}{z}\right)$$

$$\boldsymbol{p_{k+r}}[2i+1] = cos\left(\frac{k+r}{z}\right) = cos\left(\frac{k}{z}\right)cos\left(\frac{r}{z}\right) - sin\left(\frac{k}{z}\right)sin\left(\frac{r}{z}\right)$$

$$\boldsymbol{p_k}[2i] = sin\left(\frac{k}{z}\right) \qquad = \boldsymbol{p_k}[2i+1]cos\left(\frac{r}{z}\right) - \boldsymbol{p_k}[2i]sin\left(\frac{r}{z}\right)$$

$$\boldsymbol{p_k}[2i+1] = cos\left(\frac{k}{z}\right)$$

$$\boldsymbol{p_{k+r}} = M_r \boldsymbol{p_k} \qquad \boldsymbol{p_k}[2i] = sin\left(\frac{k}{10000^{2i/d}}\right) \qquad \boldsymbol{p_k}[2i+1] = cos\left(\frac{k}{10000^{2i/d}}\right)$$

z z

$$\boldsymbol{p_{k+r}}[2i] = \boldsymbol{p_k}[2i]cos\left(\frac{r}{z}\right) + \boldsymbol{p_k}[2i+1]sin\left(\frac{r}{z}\right)$$

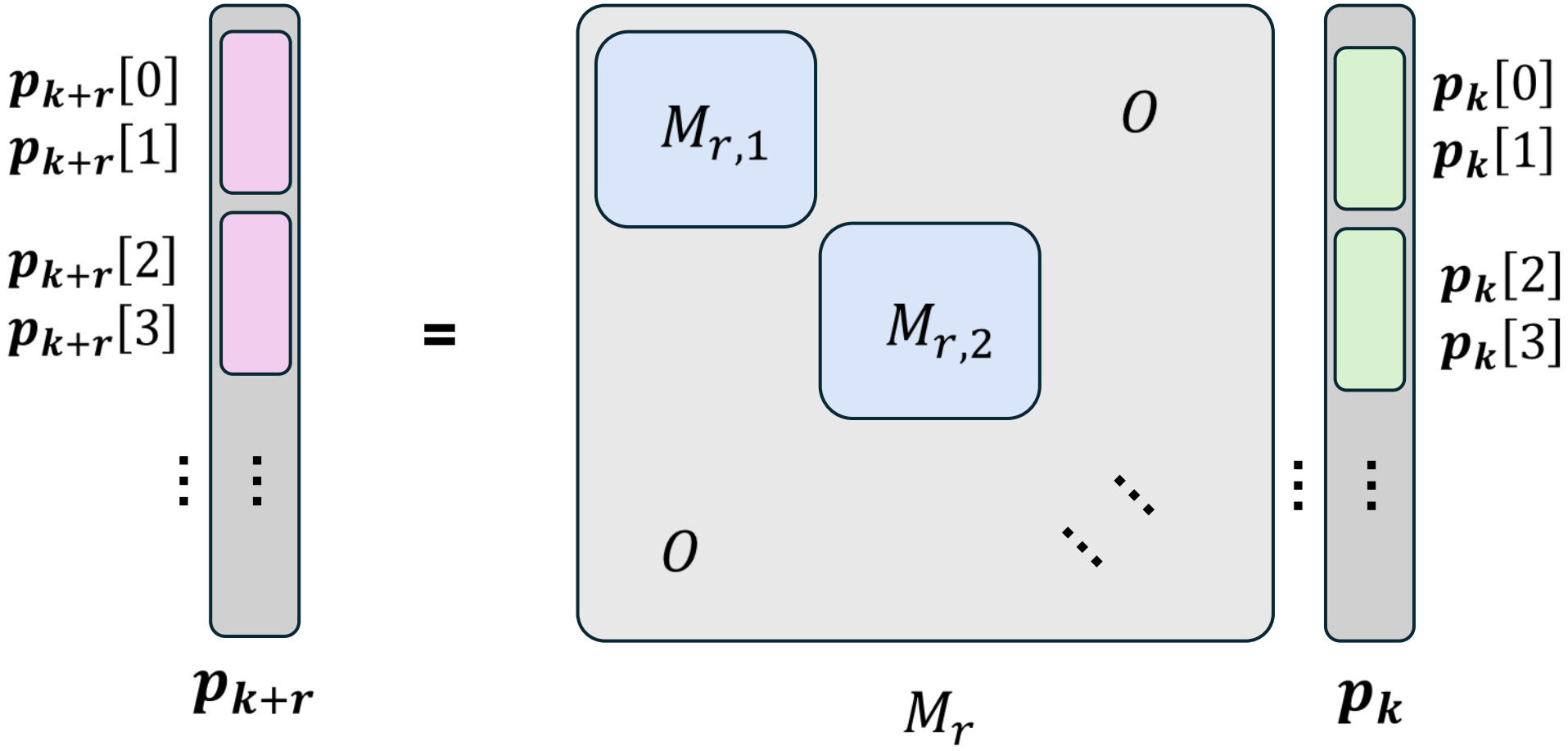
$$\boldsymbol{p_{k+r}}[2i+1] = \boldsymbol{p_k}[2i+1]cos\left(\frac{r}{z}\right) - \boldsymbol{p_k}[2i]sin\left(\frac{r}{z}\right)$$

$$\begin{bmatrix} \boldsymbol{p_{k+r}}[2i] \\ \boldsymbol{p_{k+r}}[2i+1] \end{bmatrix} = \begin{bmatrix} cos\left(\frac{r}{z}\right) & sin\left(\frac{r}{z}\right) \\ -sin\left(\frac{r}{z}\right) & cos\left(\frac{r}{z}\right) \end{bmatrix} \begin{bmatrix} \boldsymbol{p_k}[2i] \\ \boldsymbol{p_k}[2i+1] \end{bmatrix}$$

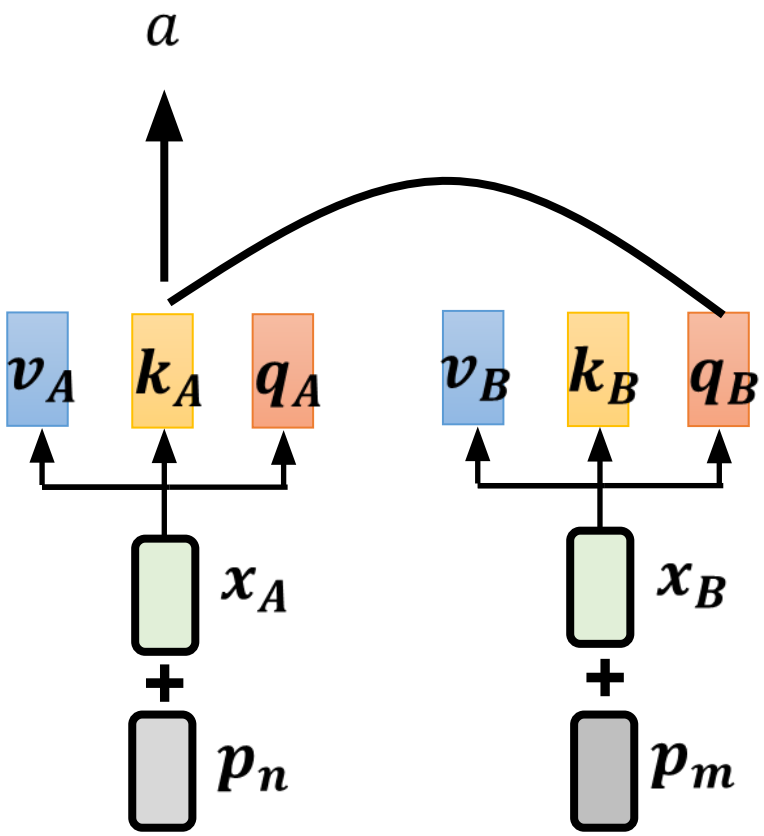
$M_{r,i}$

$$\boldsymbol{p_{k+r}} = M_r \boldsymbol{p_k} \qquad \boldsymbol{p_k}[2i] = sin\left(\frac{k}{10000^{2i/d}}\right) \qquad \boldsymbol{p_k}[2i+1] = cos\left(\frac{k}{10000^{2i/d}}\right)$$

$$\begin{bmatrix} \boldsymbol{p_{k+r}}[2i] \\ \boldsymbol{p_{k+r}}[2i+1] \end{bmatrix} = M_{r,i} \begin{bmatrix} \boldsymbol{p_k}[2i] \\ \boldsymbol{p_k}[2i+1] \end{bmatrix}$$



$$\boldsymbol{p}_{k+r} = M_r \boldsymbol{p}_k$$



$$a = \boldsymbol{q}_B \cdot \boldsymbol{k}_A \quad = (\boldsymbol{q}_B)^T \boldsymbol{k}_A$$

$$= \left(W_q(\boldsymbol{x}_B + \boldsymbol{p}_m)\right)^T W_k(\boldsymbol{x}_A + \boldsymbol{p}_n)$$

$$= (\boldsymbol{x}_B + \boldsymbol{p}_m)^T W_q{}^T W_k(\boldsymbol{x}_A + \boldsymbol{p}_n)$$

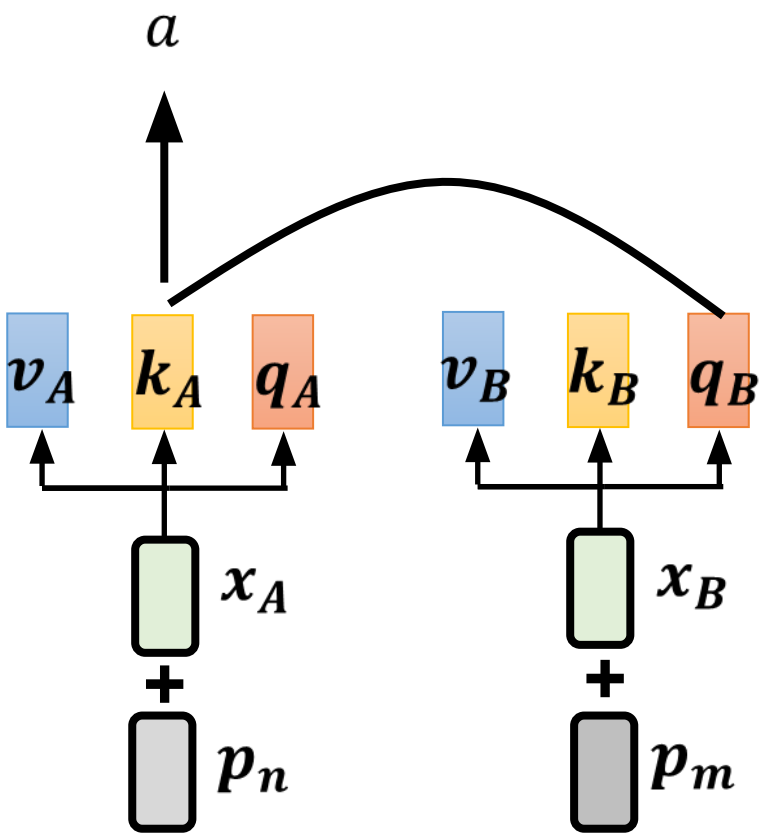
$$= \boldsymbol{x}_B{}^T W_q{}^T W_k \boldsymbol{x}_A$$ 只跟內容有關

$$+\boldsymbol{x}_B{}^T W_q{}^T W_k \boldsymbol{p}_n + \boldsymbol{p}_m{}^T W_q{}^T W_k \boldsymbol{x}_A$$ 內容、位置交互影響

$$+\boldsymbol{p}_m{}^T W_q{}^T W_k \boldsymbol{p}_n$$ 只跟位置有關

但看不出和相關位置的直接關聯

$$\boldsymbol{p}_{k+r} = M_r \boldsymbol{p}_k$$



$$a = \boldsymbol{q_B} \cdot \boldsymbol{k_A} \quad = (\boldsymbol{q_B})^T \boldsymbol{k_A}$$

$$= \left(W_q(\boldsymbol{x_B} + \boldsymbol{p_m})\right)^T W_k(\boldsymbol{x_A} + \boldsymbol{p_n})$$

$$= (\boldsymbol{x_B} + \boldsymbol{p_m})^T {W_q}^T W_k(\boldsymbol{x_A} + \boldsymbol{p_n})$$

$$= \boldsymbol{x_B}^T {W_q}^T W_k \boldsymbol{x_A} \quad \text{只跟內容有關}$$

$$+\boldsymbol{x}_\mathrm{B}{}^T {W_q}^T W_k \boldsymbol{p_n} + \boldsymbol{p_m}^T {W_q}^T W_k \boldsymbol{x_A}$$

內容、位置交互影響

$$+\boldsymbol{p_m}^T {W_q}^T W_k \boldsymbol{p_n} \quad \text{只跟位置有關}$$

$$= (M_{m-n}\boldsymbol{p_n})^T {W_q}^T W_k \boldsymbol{p_n} \quad = (\boldsymbol{p_n})^T M_{m-n} {W_q}^T W_k \boldsymbol{p_n}$$

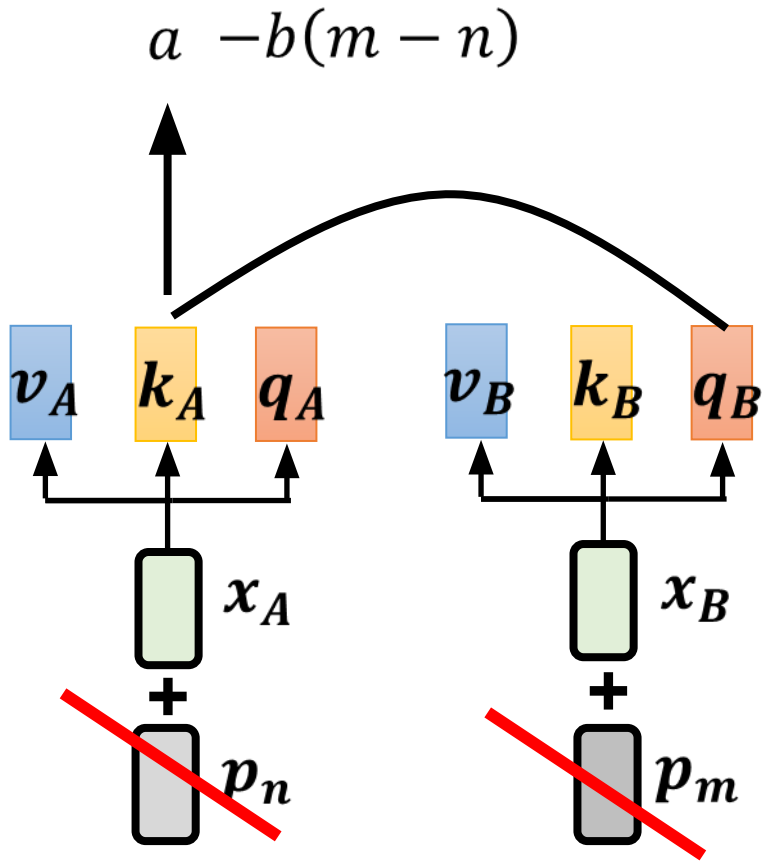
只跟相對位置有關



Relative Positional Embedding

Attention with Linear Biases (ALiBi)

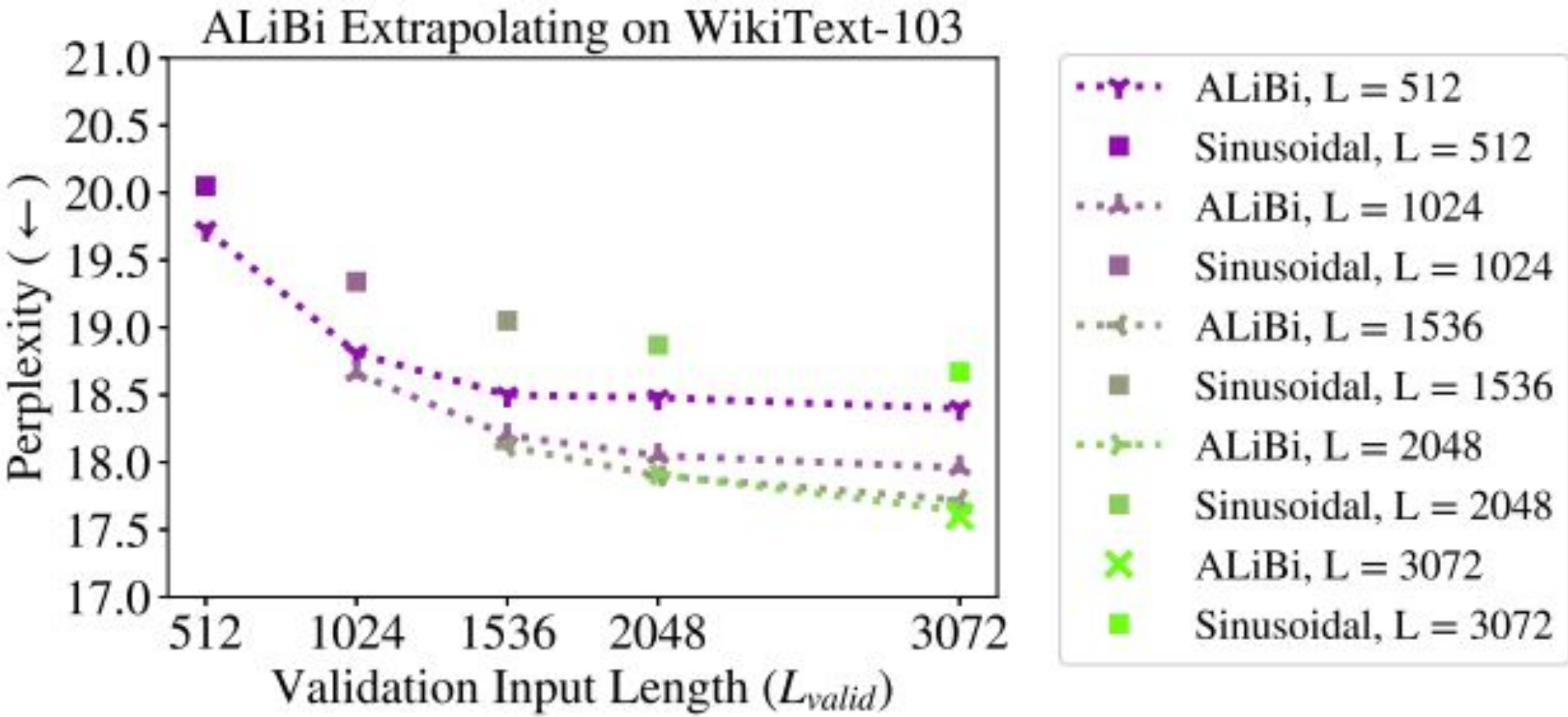
https://arxiv.org/abs/2108.12409



距離越遠，attention 越小，
就這樣!

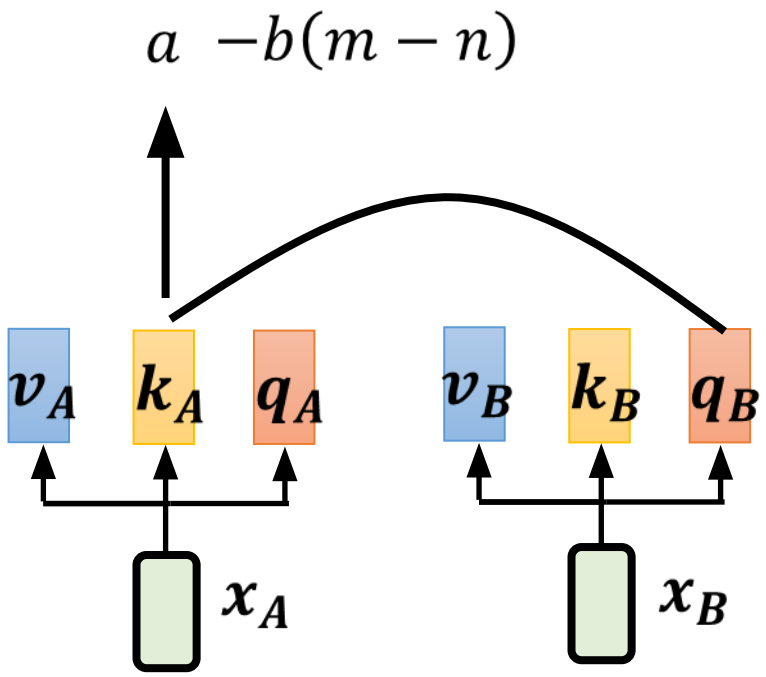
b 是手動設置

不同 attention head 設置不同值

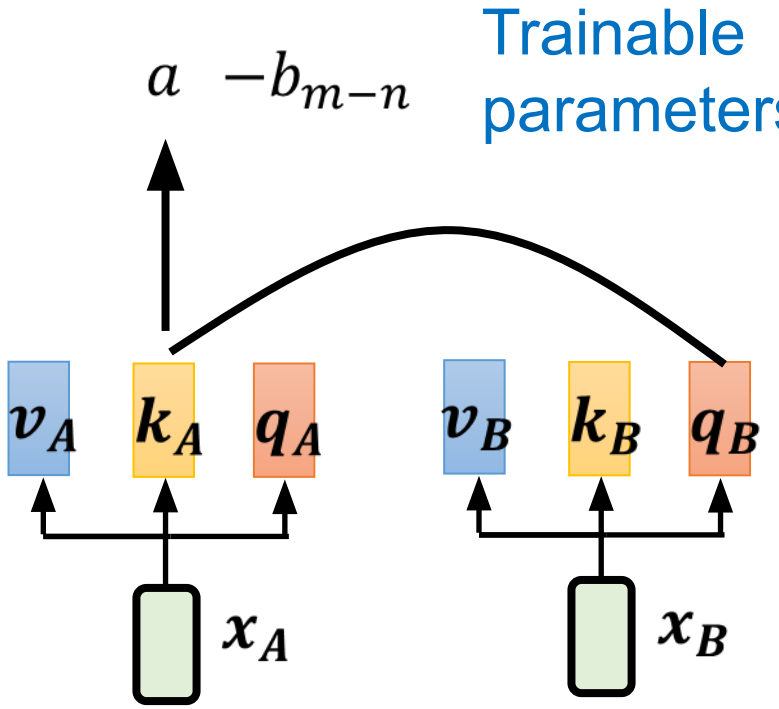


https://arxiv.org/abs/2108.12409

Text-to-Text Transfer Transformer (T5)



https://arxiv.org/abs/2108.12409



https://arxiv.org/abs/1910.10683

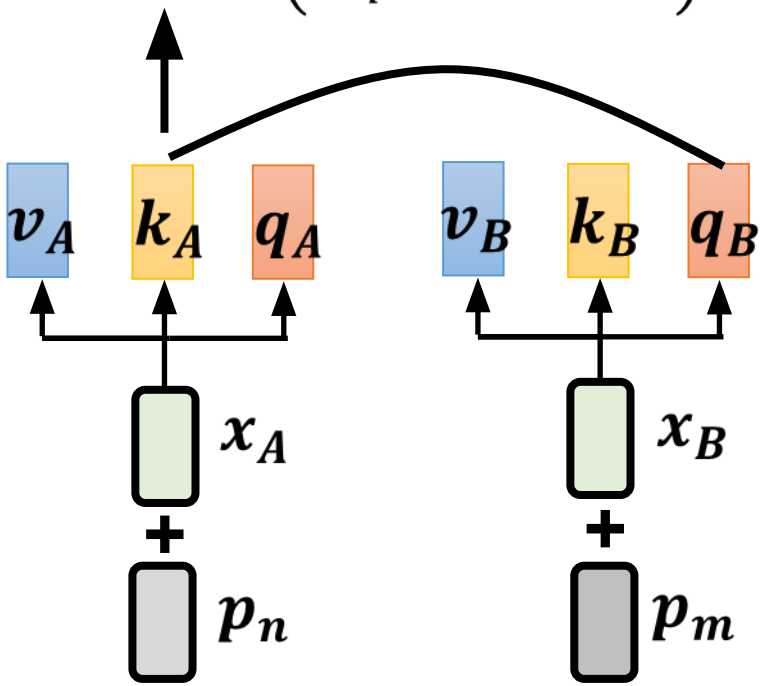
RoPE

Rotary Position Embedding

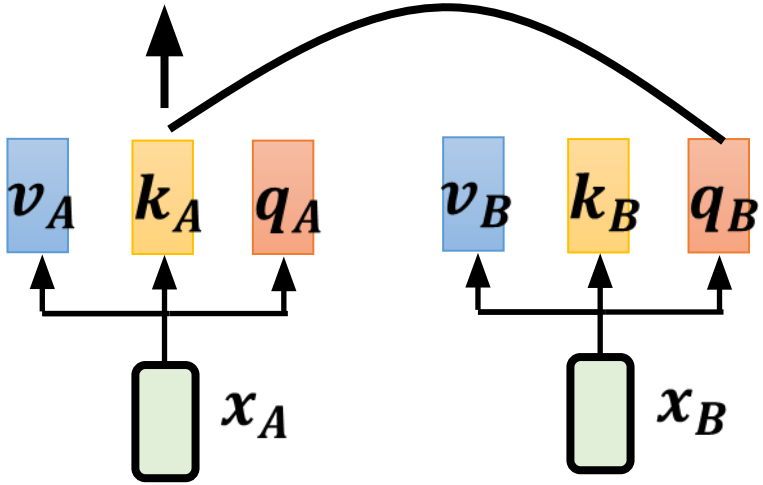
Llama、Qwen、Gemma

https://arxiv.org/abs/2104.09864

$$a = \left(W_q(x_A + p_m)\right)^T W_k(x_B + p_n)$$



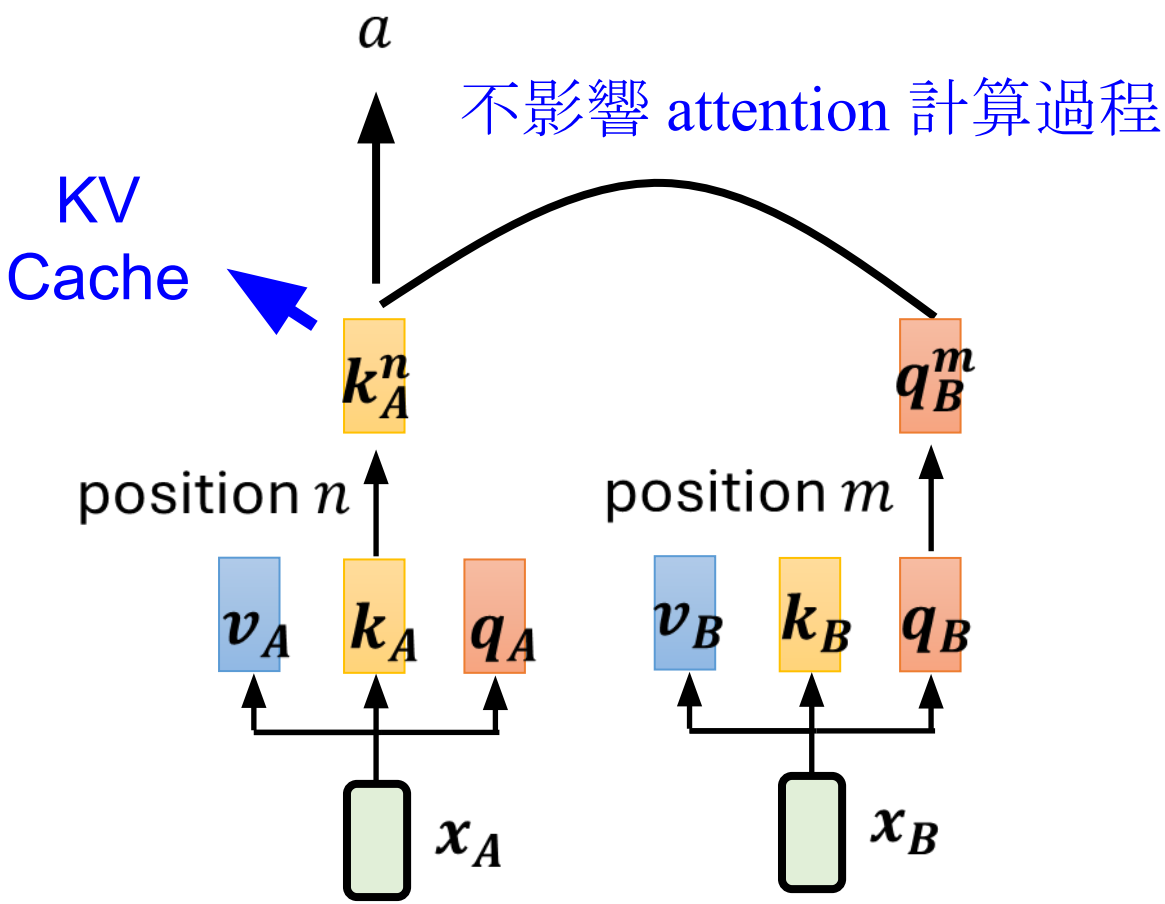
$$a - b(m-n)$$



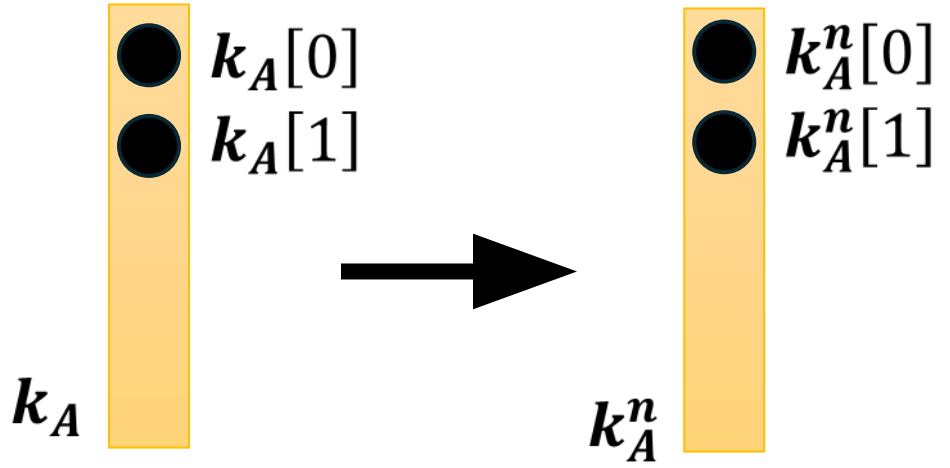
$$a = (k_A^n)^T q_B^m$$

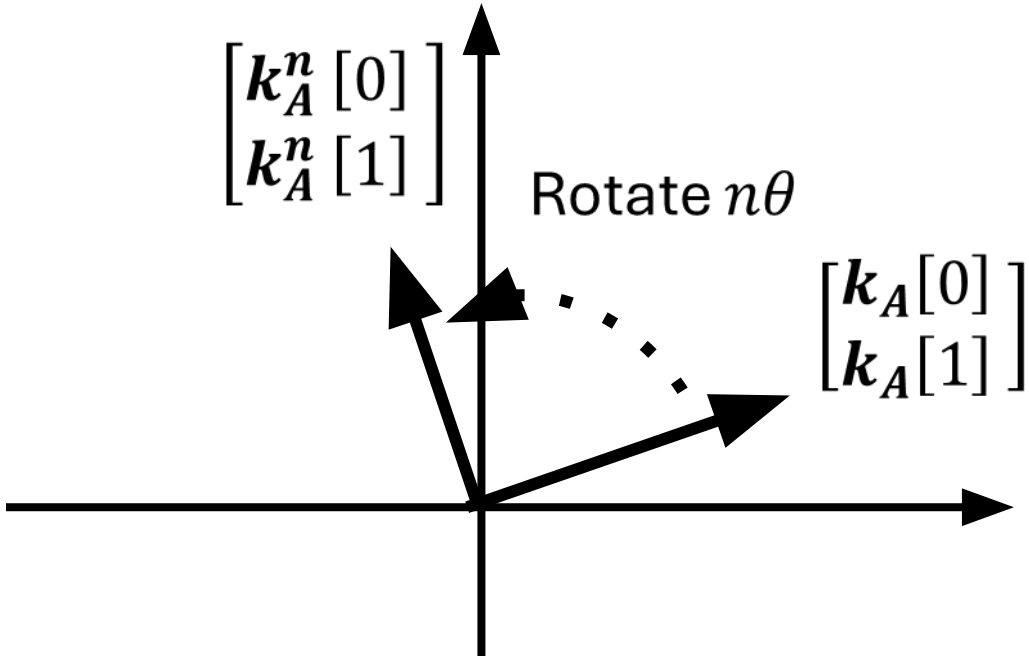
$$= (k_A)^T R_{m-n} q_B$$

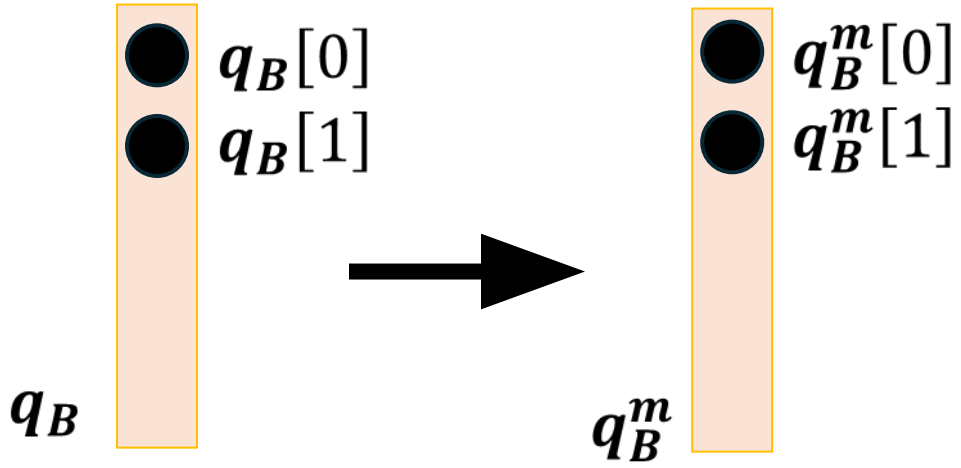
不影響 attention 計算過程

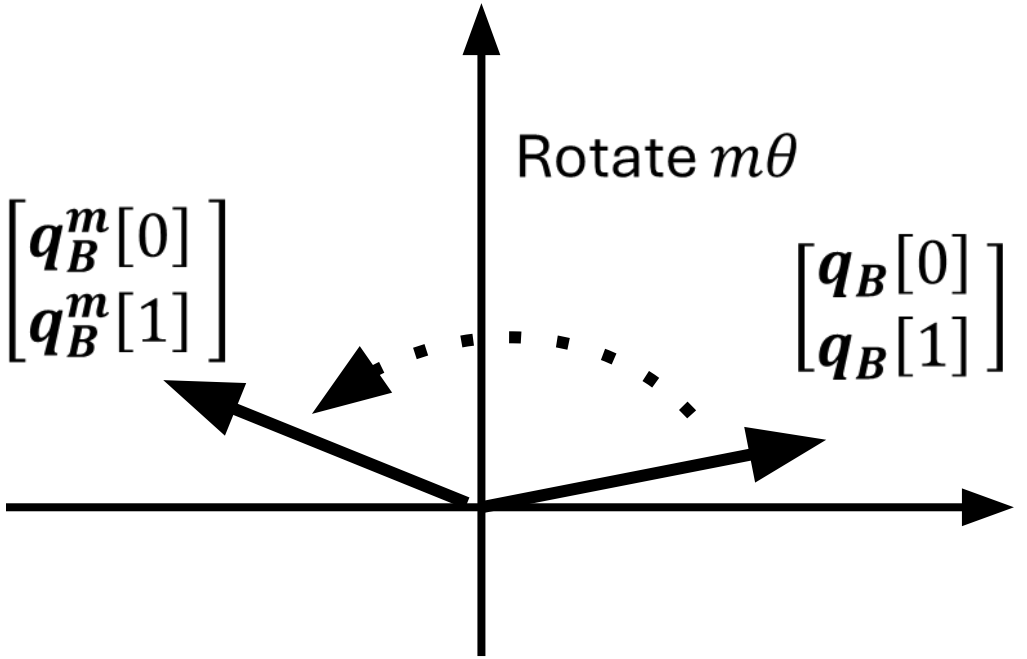


Rotation as Position

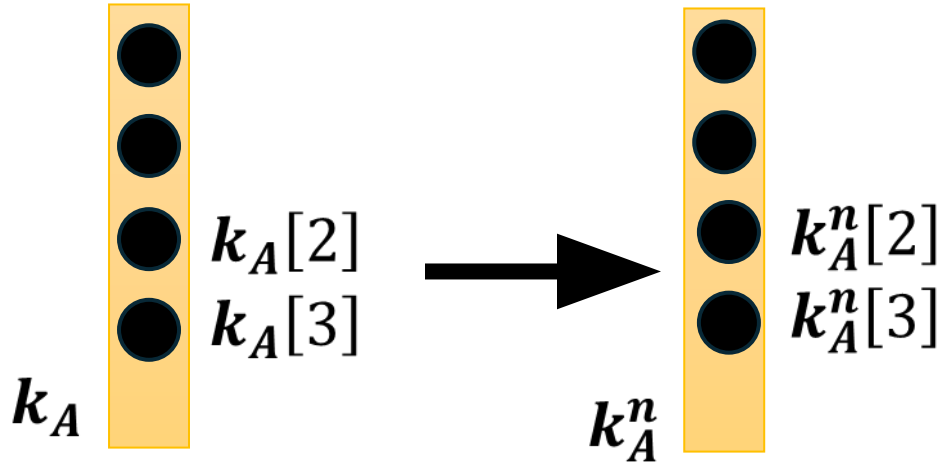


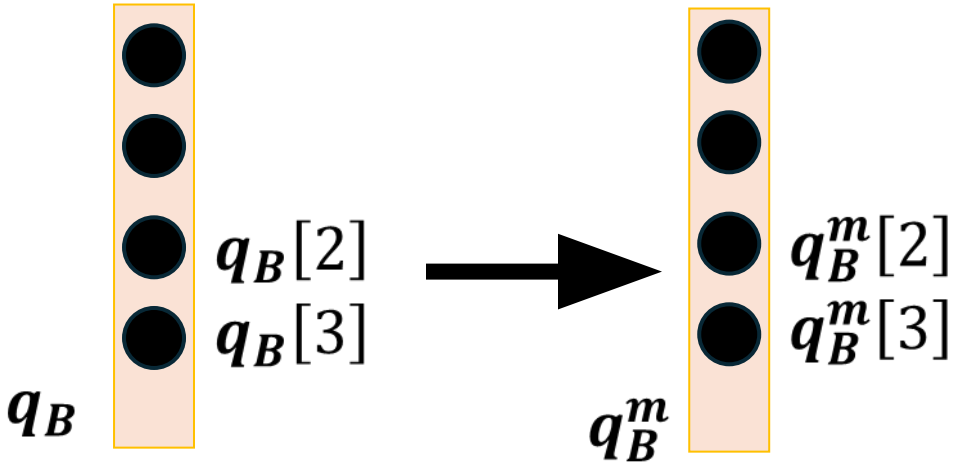


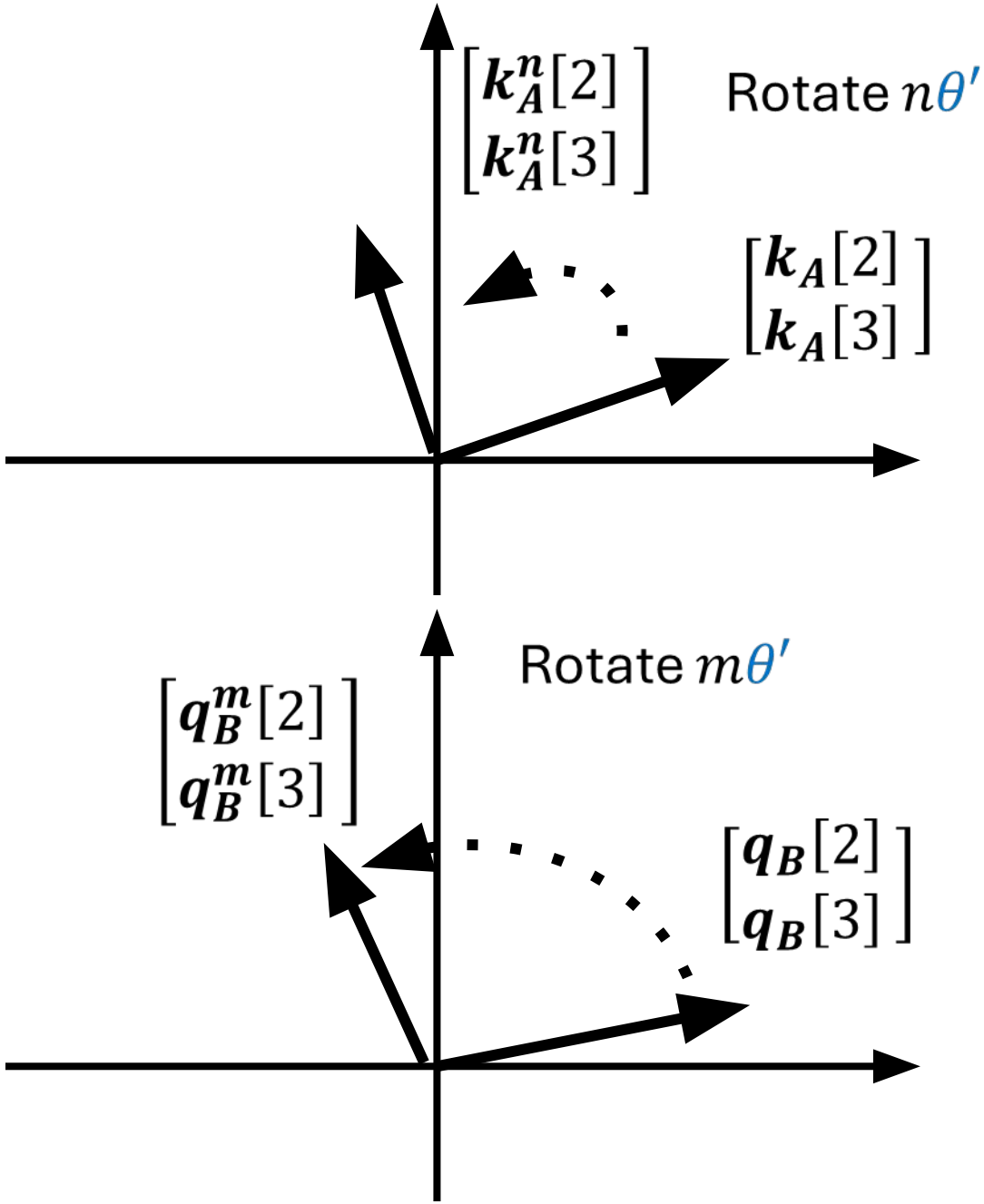




Rotation as Position

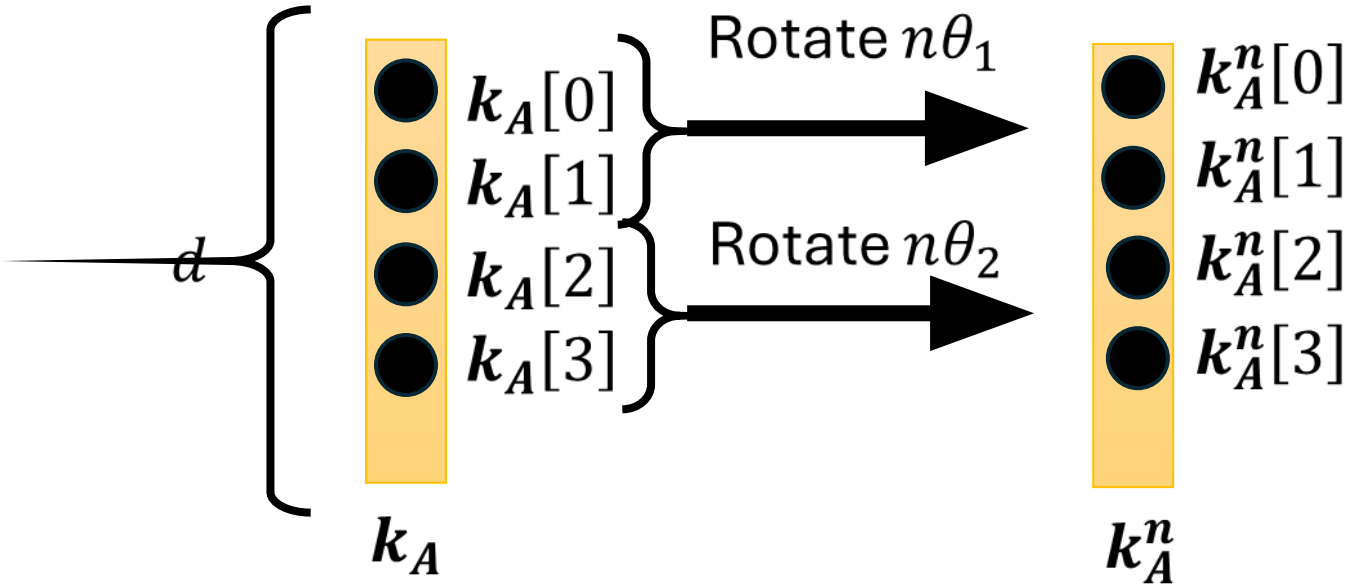






旋轉角度怎麼設定

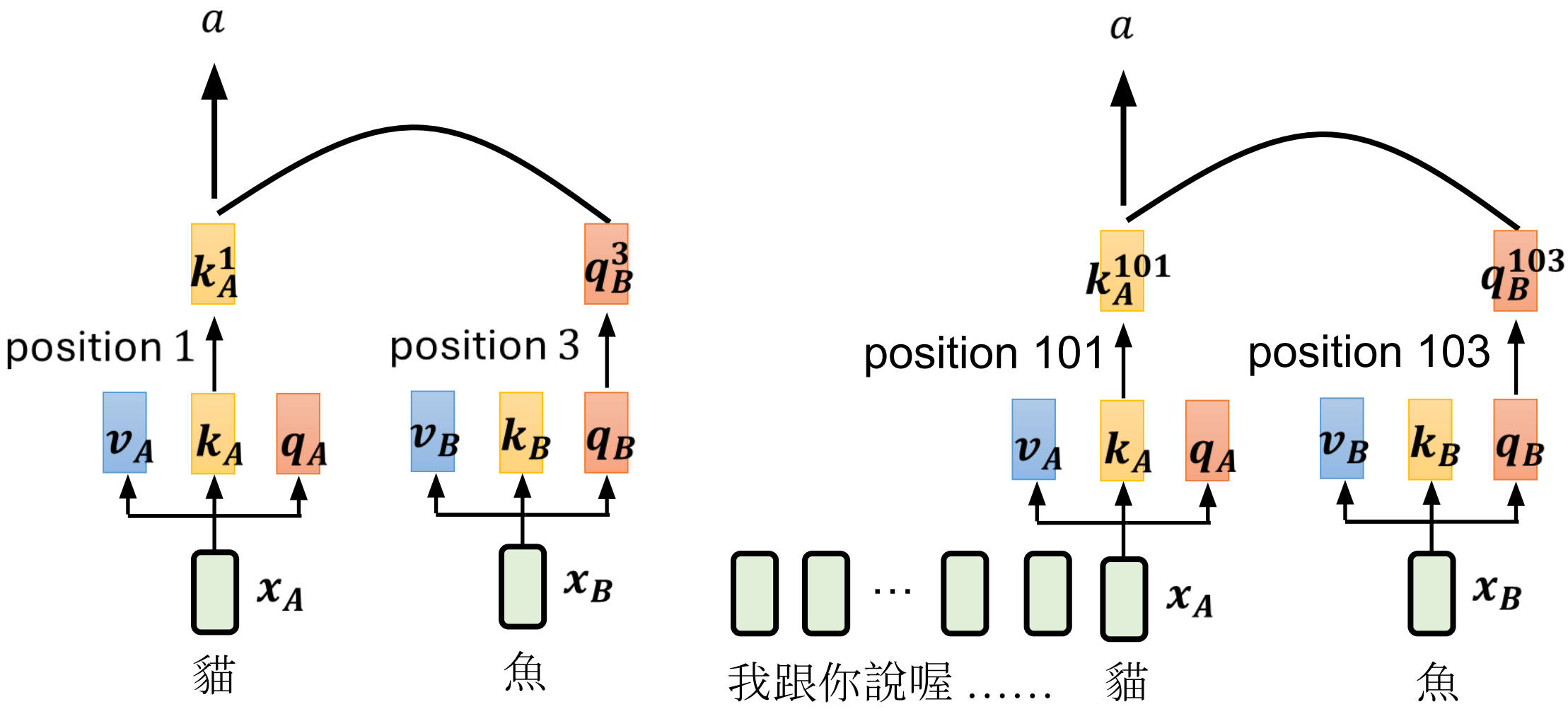
https://arxiv.org/abs/2104.09864



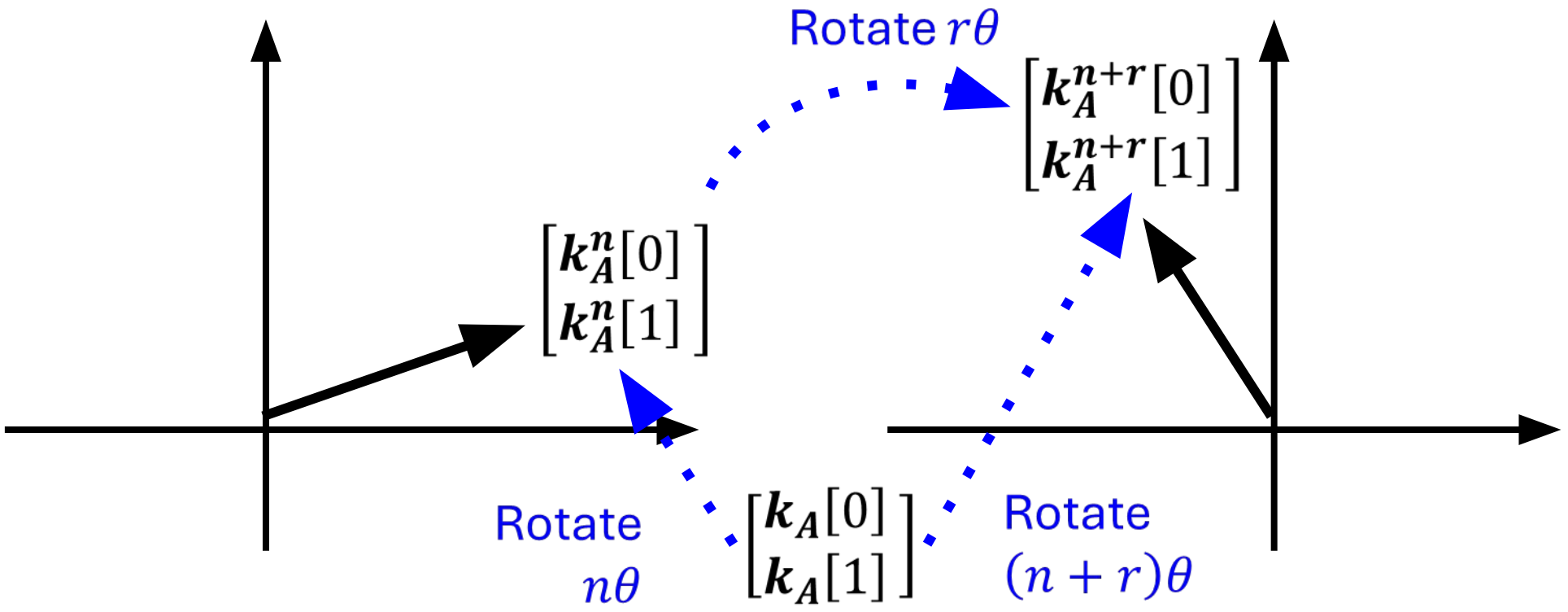
$$\theta_i = \frac{1}{10000^{2i/d}}$$

$$i = 0,1,\ldots,\frac{d}{2} - 1$$

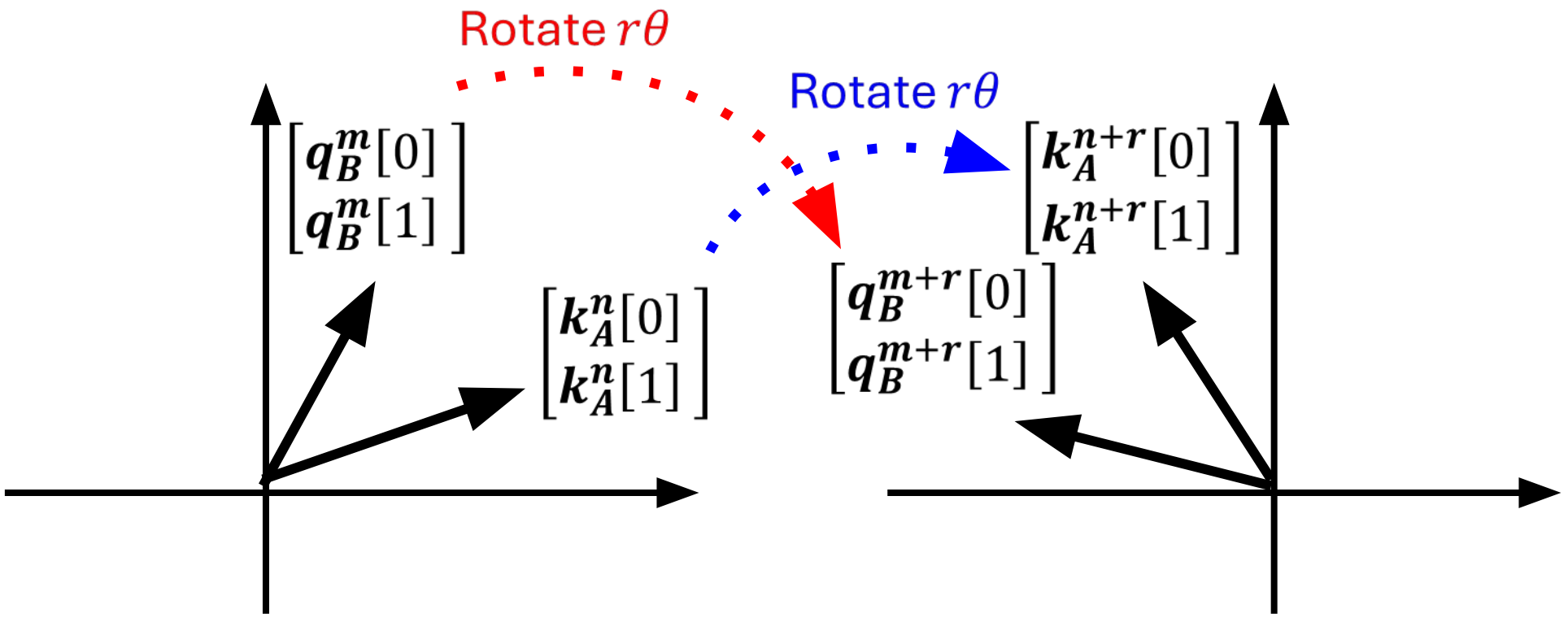
$$k_A^n \cdot q_B^m = k_A^{n+r} \cdot q_B^{m+r}$$

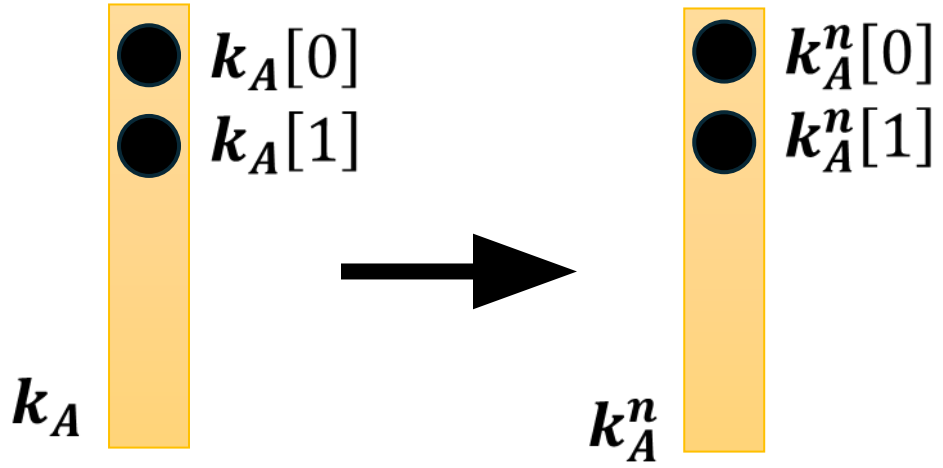


$$k_A^n \cdot q_B^m = k_A^{n+r} \cdot q_B^{m+r}$$



$$k_A^n \cdot q_B^m = k_A^{n+r} \cdot q_B^{m+r}$$



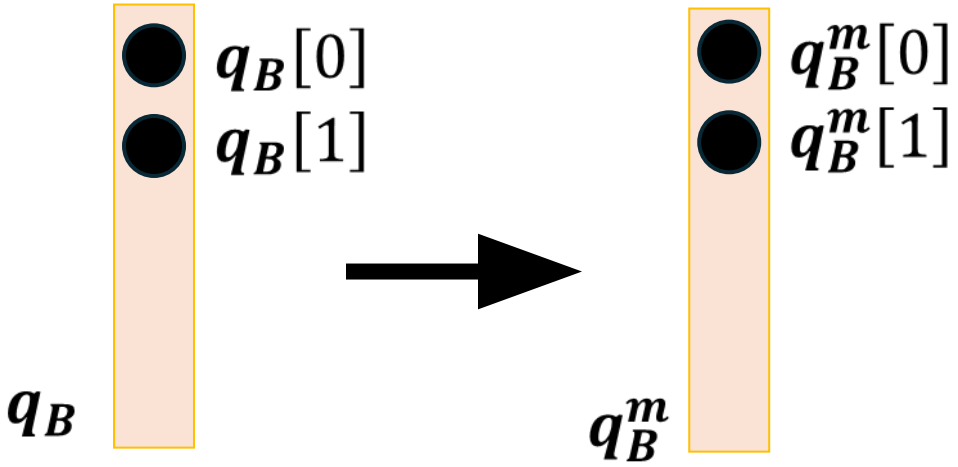


$$a = (k_A^n)^T q_B^m$$

$$= (k_A)^T R_{m-n} q_B$$

Rotate $n\theta$

$$\begin{bmatrix} k_A^n[0] \\ k_A^n[1] \end{bmatrix} = \begin{bmatrix} cos(n\theta) & -sin(n\theta) \\ sin(n\theta) & cos(n\theta) \end{bmatrix} \begin{bmatrix} k_A[0] \\ k_A[1] \end{bmatrix}$$



Rotate $m\theta$

$$\begin{bmatrix} q_B^m[0] \\ q_B^m[1] \end{bmatrix} = \begin{bmatrix} cos(m\theta) & -sin(m\theta) \\ sin(m\theta) & cos(m\theta) \end{bmatrix} \begin{bmatrix} q_B[0] \\ q_B[1] \end{bmatrix}$$

Rotate $n\theta$

$$\begin{bmatrix} \boldsymbol{k_A^n}[0] \\ \boldsymbol{k_A^n}[1] \end{bmatrix} = \begin{bmatrix} cos(n\theta) & -sin(n\theta) \\ sin(n\theta) & cos(n\theta) \end{bmatrix} \begin{bmatrix} \boldsymbol{k_A}[0] \\ \boldsymbol{k_A}[1] \end{bmatrix}$$

Rotate $m\theta$

$$\begin{bmatrix} \boldsymbol{q_B^m}[0] \\ \boldsymbol{q_B^m}[1] \end{bmatrix} = \begin{bmatrix} cos(m\theta) & -sin(m\theta) \\ sin(m\theta) & cos(m\theta) \end{bmatrix} \begin{bmatrix} \boldsymbol{q_B}[0] \\ \boldsymbol{q_B}[1] \end{bmatrix}$$

$$\begin{bmatrix} \boldsymbol{k_A^n}[0] \\ \boldsymbol{k_A^n}[1] \end{bmatrix} \cdot \begin{bmatrix} \boldsymbol{q_B^m}[0] \\ \boldsymbol{q_B^m}[1] \end{bmatrix} = \begin{bmatrix} \boldsymbol{k_A^n}[0] \\ \boldsymbol{k_A^n}[1] \end{bmatrix}^T \begin{bmatrix} \boldsymbol{q_B^m}[0] \\ \boldsymbol{q_B^m}[1] \end{bmatrix}$$

$$= \left(\begin{bmatrix} cos(n\theta) & -sin(n\theta) \\ sin(n\theta) & cos(n\theta) \end{bmatrix} \begin{bmatrix} \boldsymbol{k_A}[0] \\ \boldsymbol{k_A}[1] \end{bmatrix} \right)^T \begin{bmatrix} cos(m\theta) & -sin(m\theta) \\ sin(m\theta) & cos(m\theta) \end{bmatrix} \begin{bmatrix} \boldsymbol{q_B}[0] \\ \boldsymbol{q_B}[1] \end{bmatrix}$$

$$= \begin{bmatrix} \boldsymbol{k_A}[0] \\ \boldsymbol{k_A}[1] \end{bmatrix}^T \begin{bmatrix} cos(n\theta) & -sin(n\theta) \\ sin(n\theta) & cos(n\theta) \end{bmatrix}^T \begin{bmatrix} cos(m\theta) & -sin(m\theta) \\ sin(m\theta) & cos(m\theta) \end{bmatrix} \begin{bmatrix} \boldsymbol{q_B}[0] \\ \boldsymbol{q_B}[1] \end{bmatrix}$$

Rotate $-n\theta$ Rotate $m\theta$

$$= \begin{bmatrix} \boldsymbol{k_A}[0] \\ \boldsymbol{k_A}[1] \end{bmatrix}^T \begin{bmatrix} cos\big((m-n)\theta\big) & -sin\big((m-n)\theta\big) \\ sin\big((m-n)\theta\big) & cos\big((m-n)\theta\big) \end{bmatrix} \begin{bmatrix} \boldsymbol{q_B}[0] \\ \boldsymbol{q_B}[1] \end{bmatrix}$$

Rotate$(m-n)\theta$

Rotate $n\theta$

$$\begin{bmatrix} \boldsymbol{k_A^n}[0] \\ \boldsymbol{k_A^n}[1] \end{bmatrix} = \begin{bmatrix} cos(n\theta) & -sin(n\theta) \\ sin(n\theta) & cos(n\theta) \end{bmatrix} \begin{bmatrix} \boldsymbol{k_A}[0] \\ \boldsymbol{k_A}[1] \end{bmatrix}$$

Rotate $m\theta$

$$\begin{bmatrix} \boldsymbol{q_B^m}[0] \\ \boldsymbol{q_B^m}[1] \end{bmatrix} = \begin{bmatrix} cos(m\theta) & -sin(m\theta) \\ sin(m\theta) & cos(m\theta) \end{bmatrix} \begin{bmatrix} \boldsymbol{q_B}[0] \\ \boldsymbol{q_B}[1] \end{bmatrix}$$

$$\begin{bmatrix} \boldsymbol{k_A^n}[0] \\ \boldsymbol{k_A^n}[1] \end{bmatrix} \cdot \begin{bmatrix} \boldsymbol{q_B^m}[0] \\ \boldsymbol{q_B^m}[1] \end{bmatrix} = \begin{bmatrix} \boldsymbol{k_A}[0] \\ \boldsymbol{k_A}[1] \end{bmatrix}^T \begin{bmatrix} cos\big((m-n)\theta\big) & -sin\big((m-n)\theta\big) \\ sin\big((m-n)\theta\big) & cos\big((m-n)\theta\big) \end{bmatrix} \begin{bmatrix} \boldsymbol{q_B}[0] \\ \boldsymbol{q_B}[1] \end{bmatrix}$$

Rotate $(m-n)\theta$

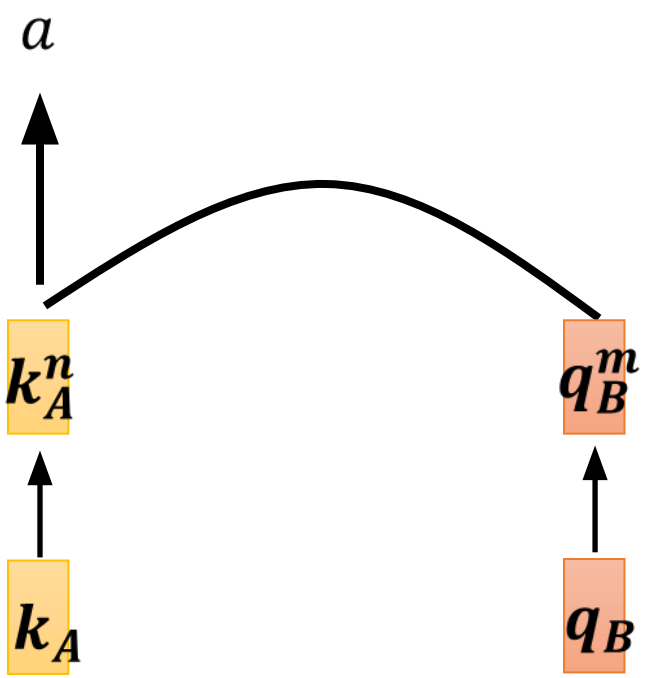
$$\begin{bmatrix} \boldsymbol{k_A^{n+r}}[0] \\ \boldsymbol{k_A^{n+r}}[1] \end{bmatrix} \cdot \begin{bmatrix} \boldsymbol{q_B^{m+r}}[0] \\ \boldsymbol{q_B^{m+r}}[1] \end{bmatrix}$$

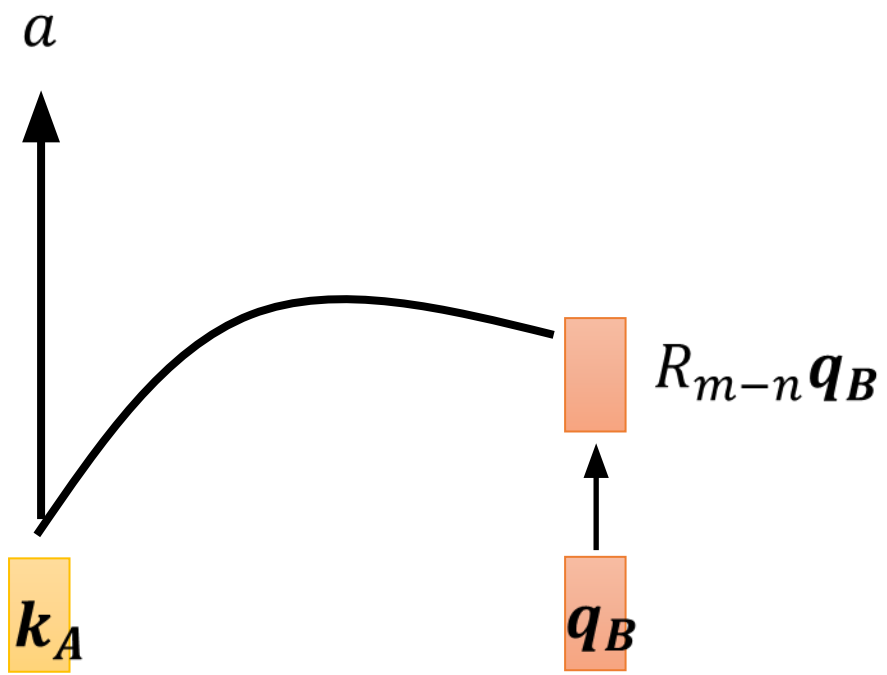
$$= \begin{bmatrix} \boldsymbol{k_A}[0] \\ \boldsymbol{k_A}[1] \end{bmatrix}^T \begin{bmatrix} cos\Big(\big((m+r)-(n+r)\big)\theta\Big) & -sin\Big(\big((m+r)-(n+r)\big)\theta\Big) \\ sin\Big(\big((m+r)-(n+r)\big)\theta\Big) & cos\Big(\big((m+r)-(n+r)\big)\theta\Big) \end{bmatrix} \begin{bmatrix} \boldsymbol{q_B}[0] \\ \boldsymbol{q_B}[1] \end{bmatrix}$$

Rotate $\big((m+r)-(n+r)\big)\theta$ = Rotate $(m-n)\theta$

$$k_A^n \cdot q_B^m = k_A^{n+r} \cdot q_B^{m+r}$$

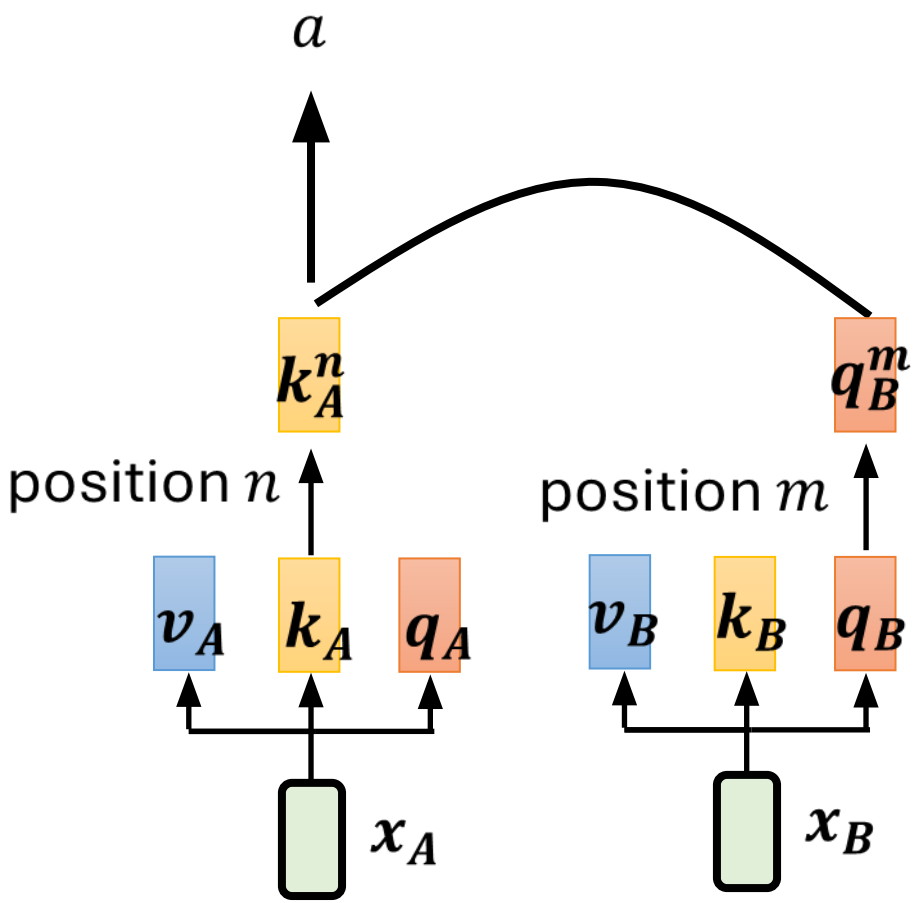
$$= (k_A)^T R_{m-n} q_B$$





RoPE 越遠越小?

https://arxiv.org/pdf/2410.06205

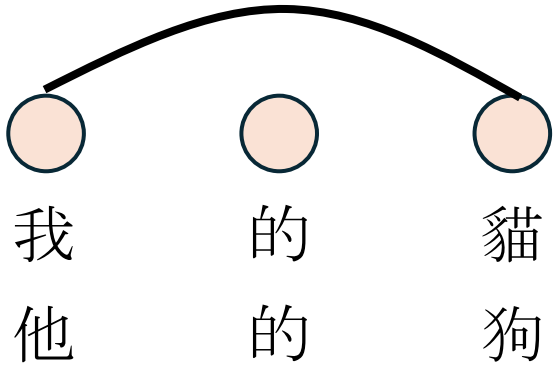


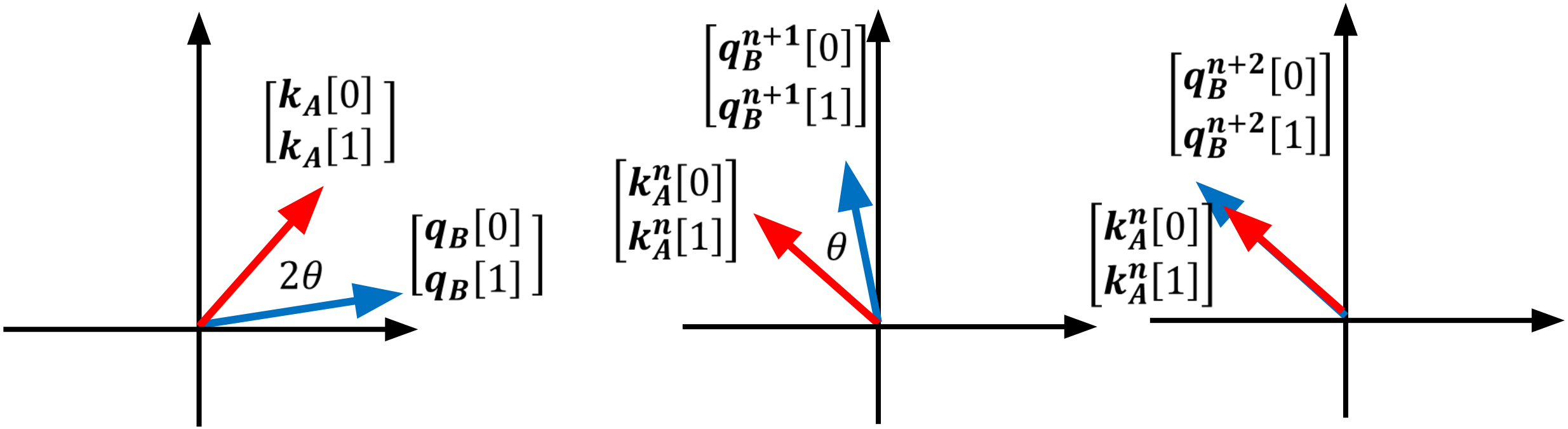
k_A 和 q_B 距離越遠

Attention weight a 算出來就越小?

https://colab.research.google.com/drive/1rWDtAkScrb2K3tcprSTzwuRQo5bGiyKJ?usp=sharing

RoPE 沒有越遠越小
不一定是一件壞事

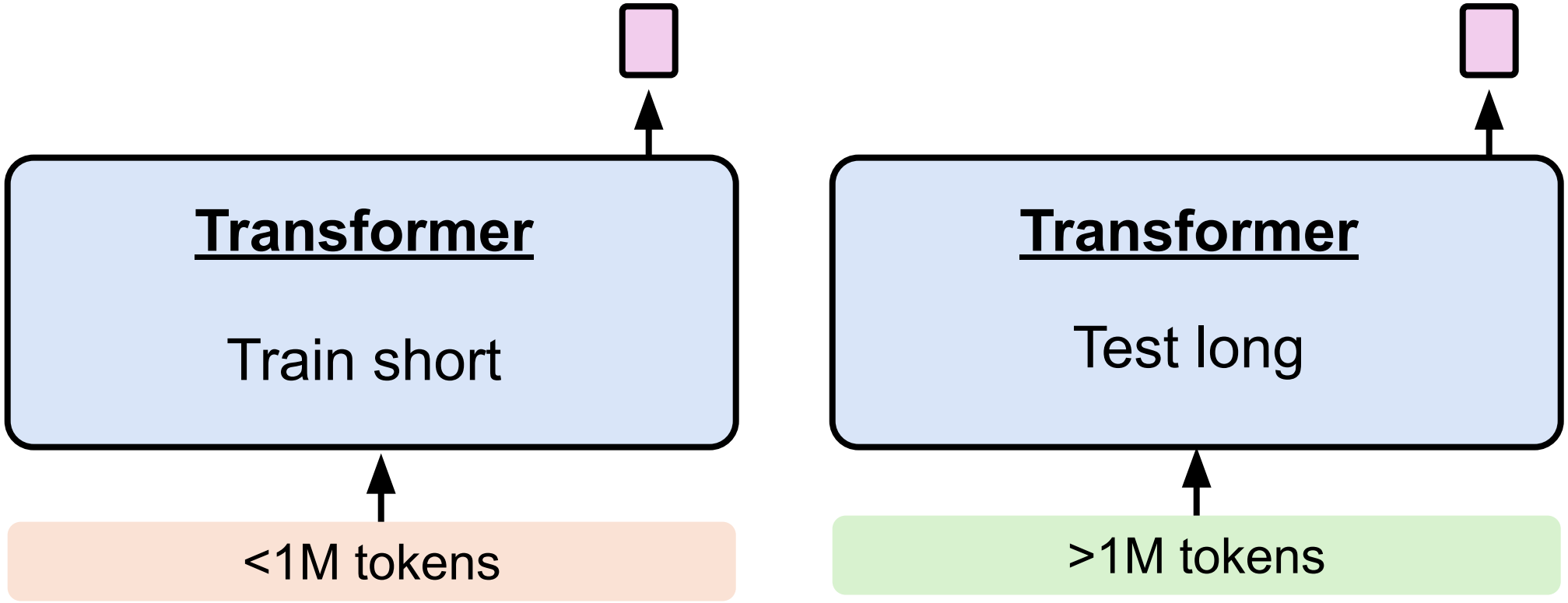




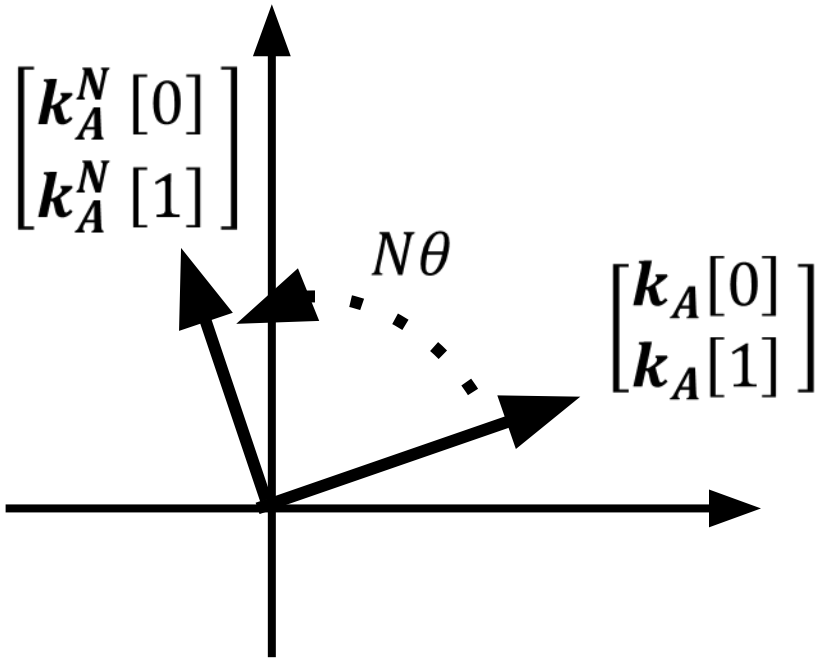
Train Short, Test Long

Reference:
https://amaarora.github.io/posts/2025-09-21-rope-context-extension.html

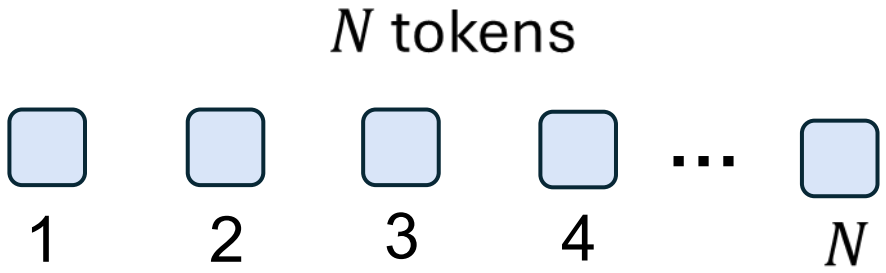
Train Short, Test Long



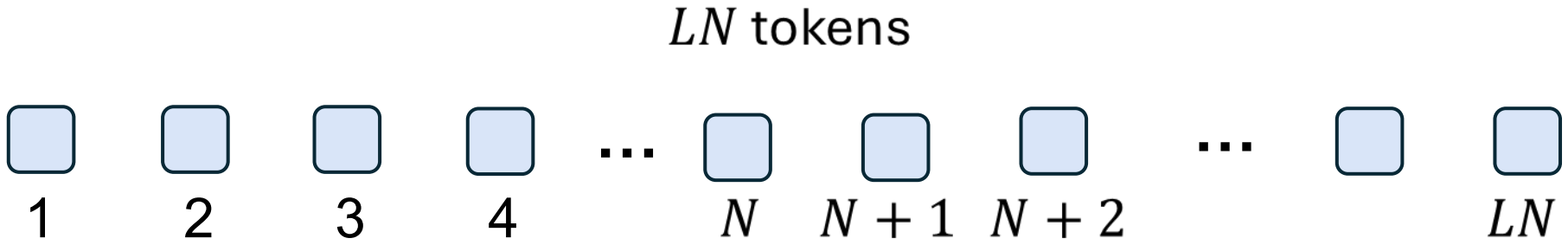
Train Short, Test Long

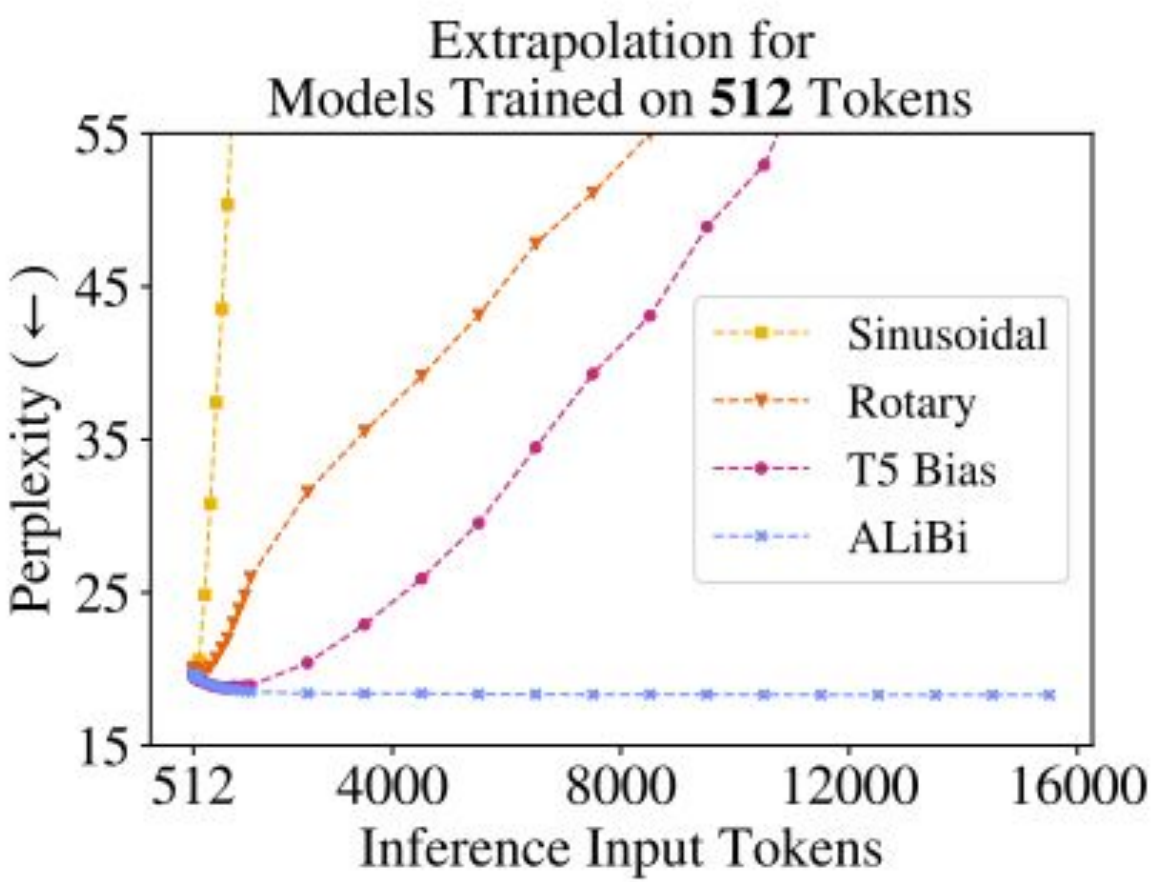


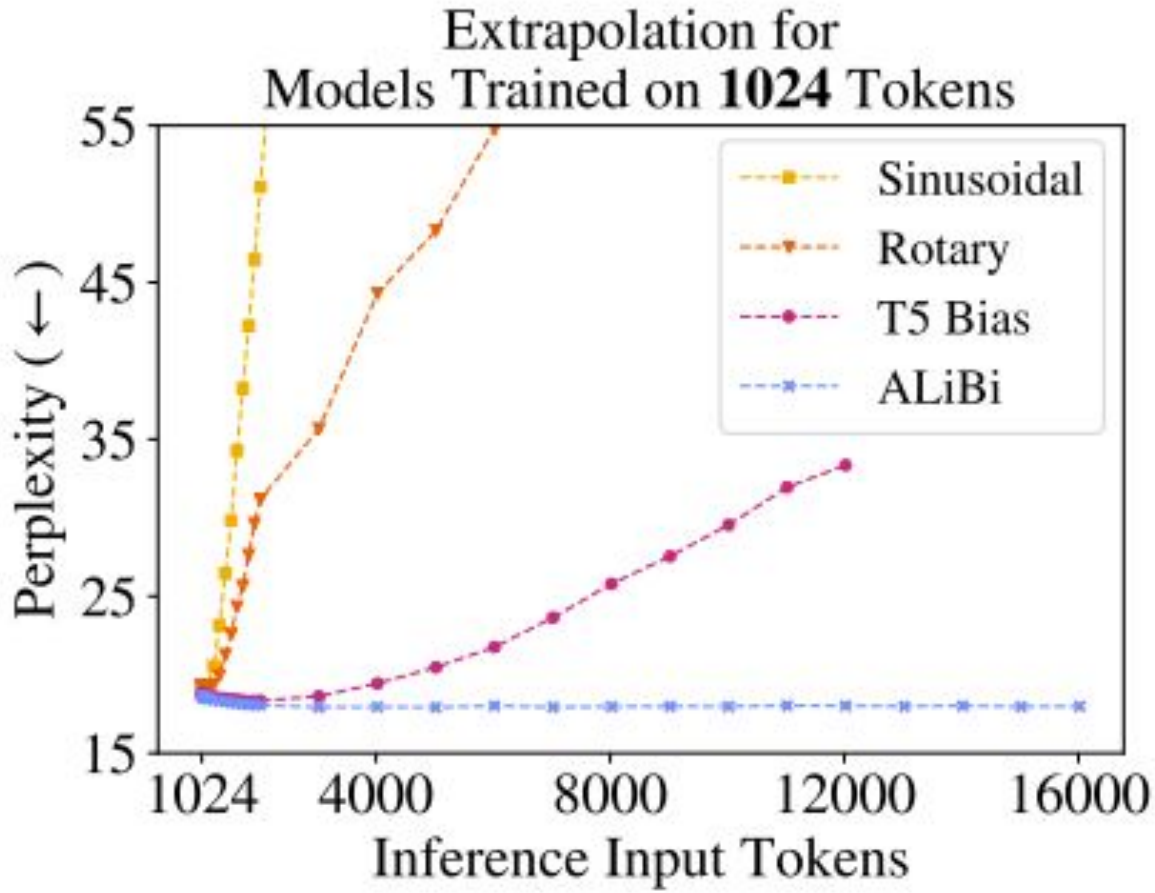
Train



Test

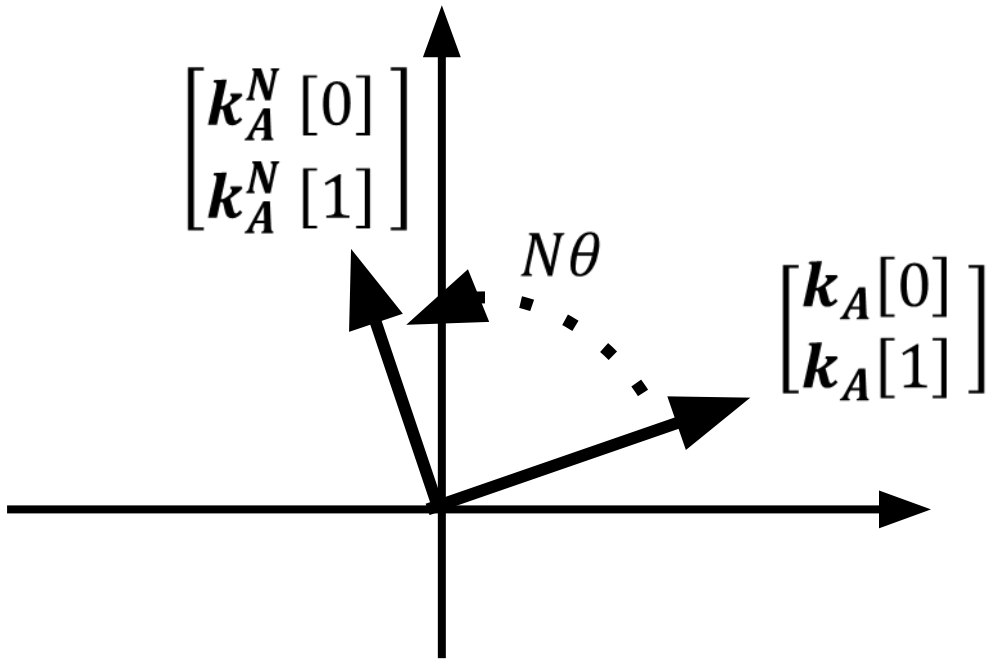




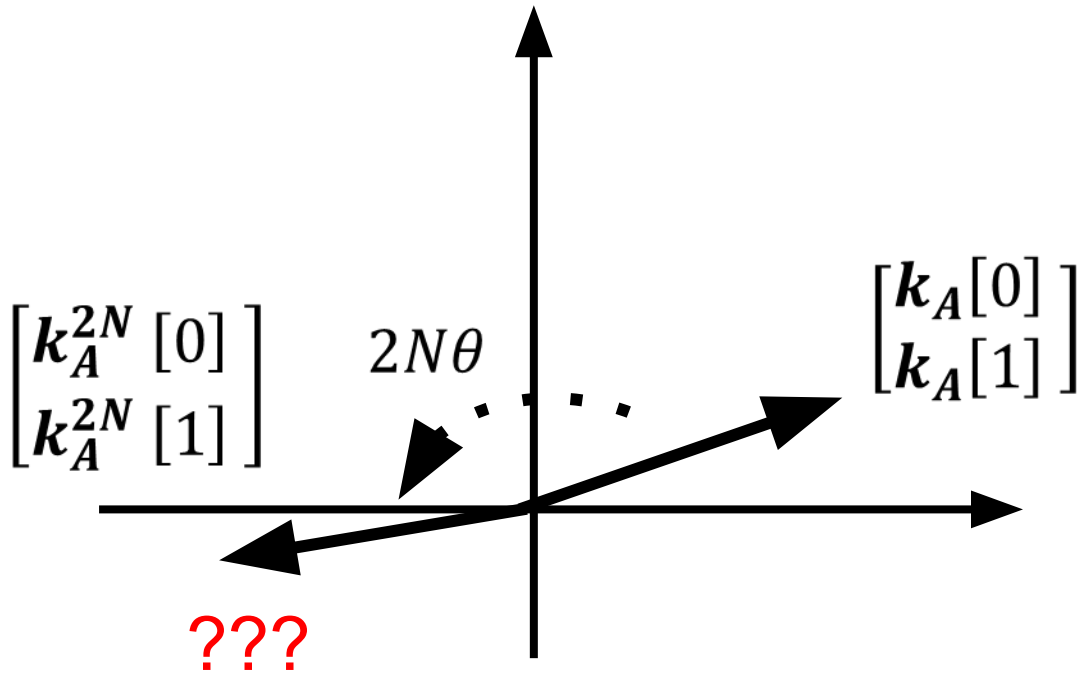


https://arxiv.org/abs/2108.12409

Why Fail?



Training: rotate at most $N\theta$

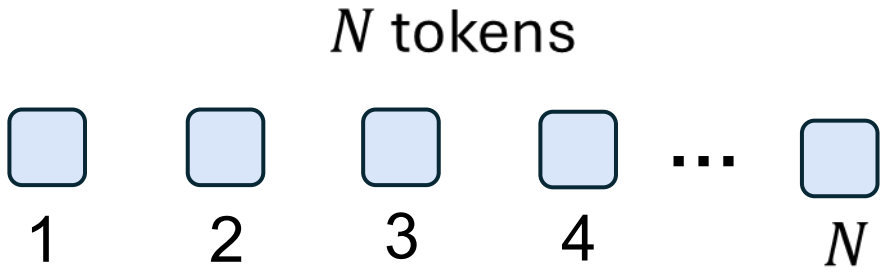


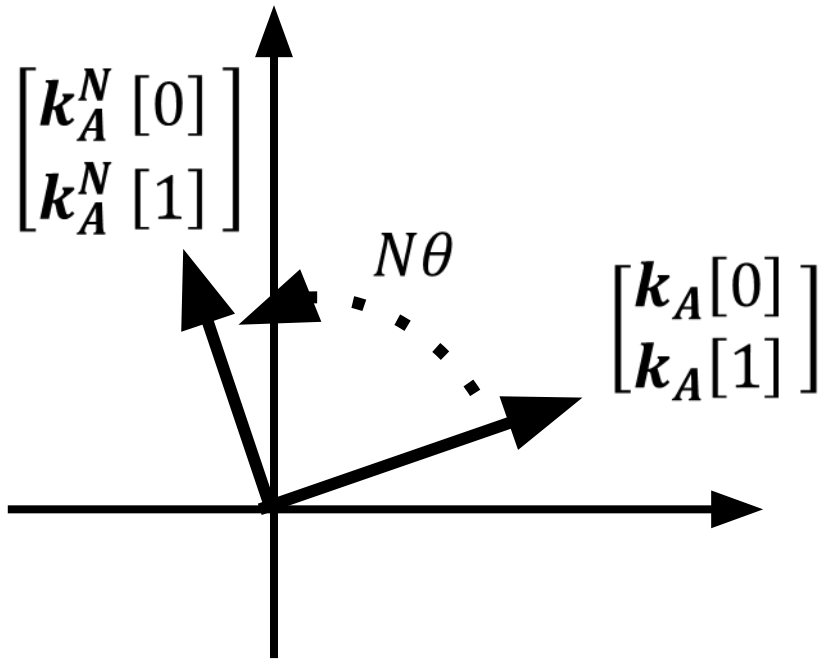
Testing: rotate $2N\theta$

https://arxiv.org/pdf/2306.15595
https://kaiokendev.github.io/context#a-bigger-problem

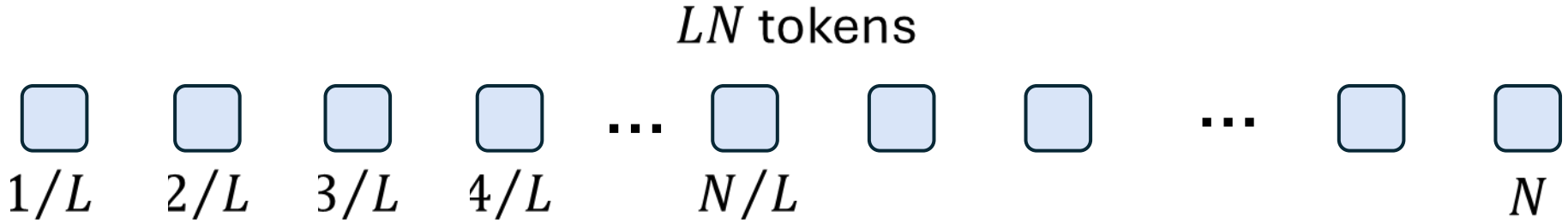
Position Interpolation

Train





Test



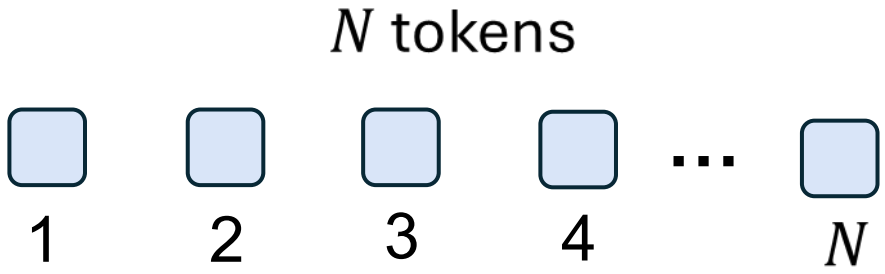
伸長 L 倍，position index 乘上 $1/L$

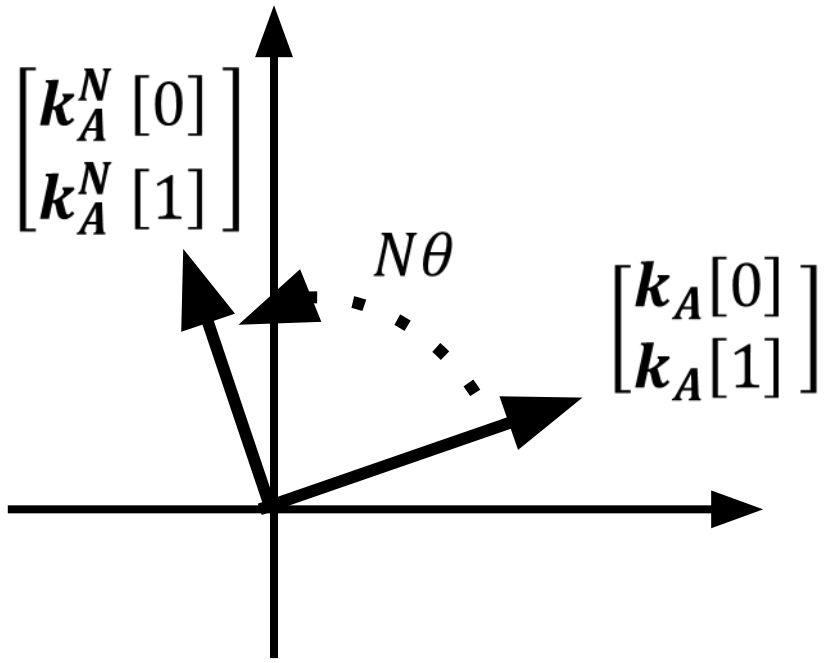
https://arxiv.org/pdf/2306.15595
https://kaiokendev.github.io/context#a-bigger-problem

Position Interpolation

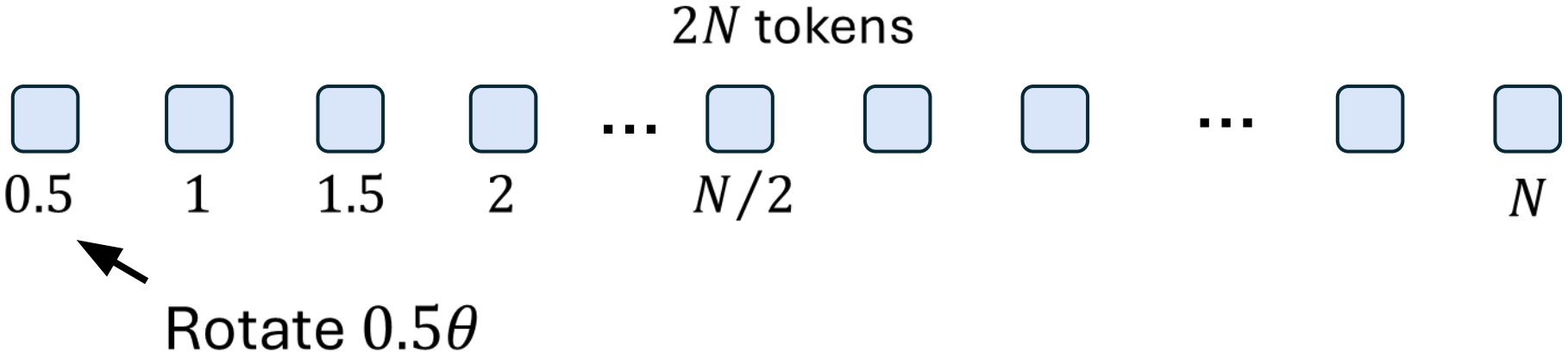
仍然需要微調模型參數

Train





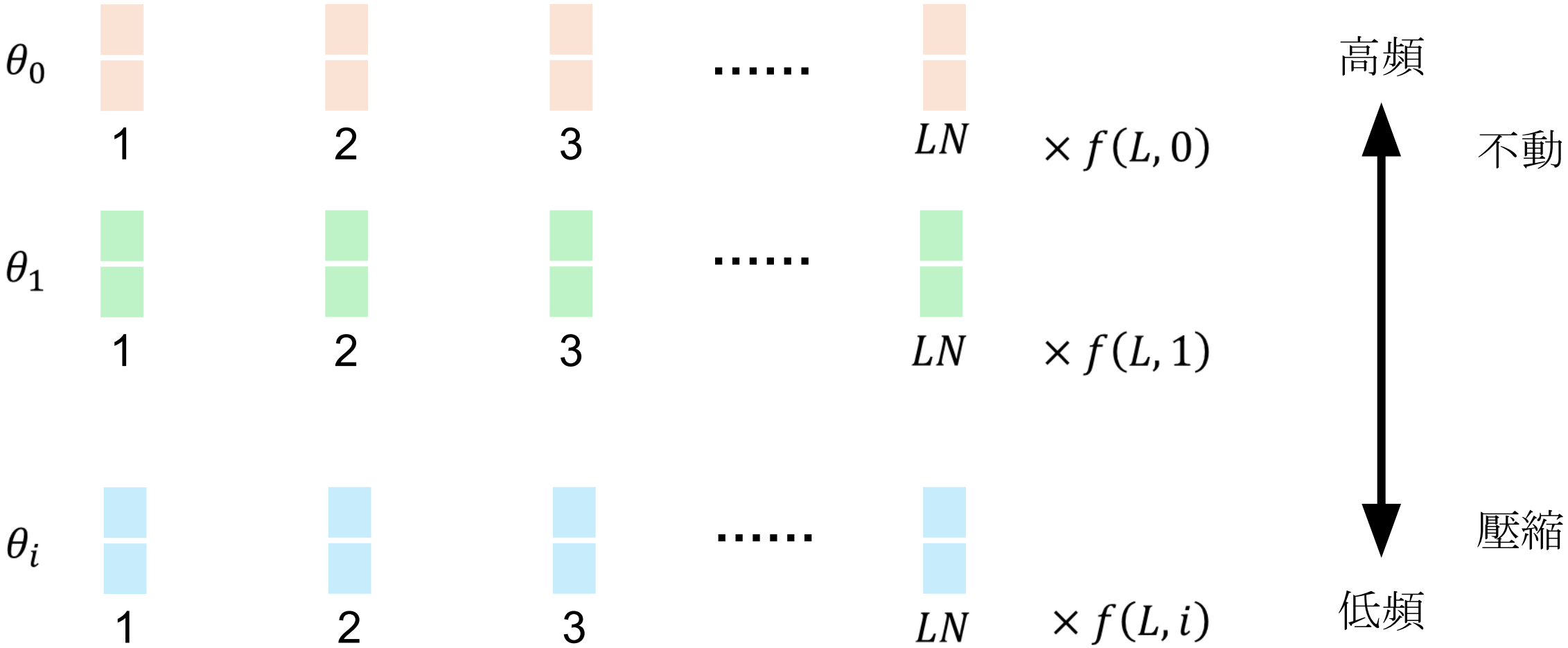
Test



Frequency-Based Approach

$$\theta_i = \frac{1}{10000^{2i/d}}$$

$$i = 0,1,\ldots,\frac{d}{2} - 1$$

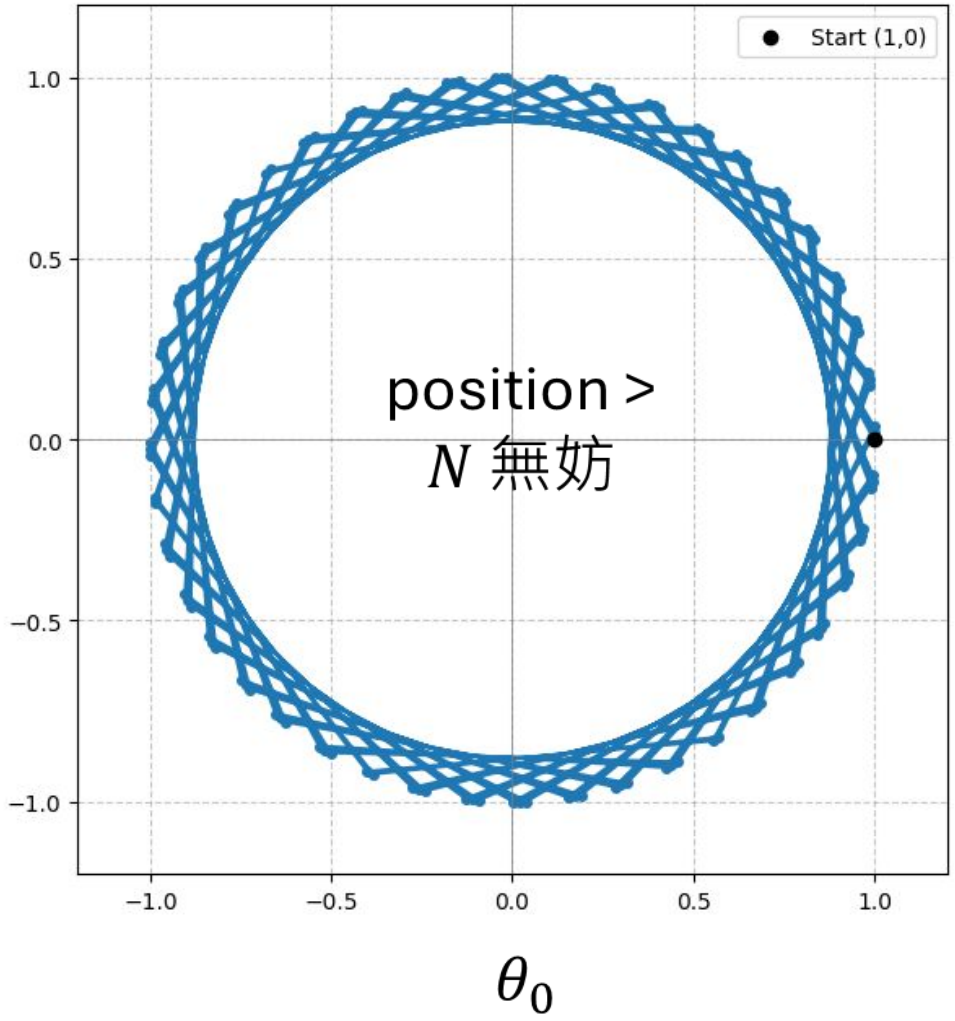


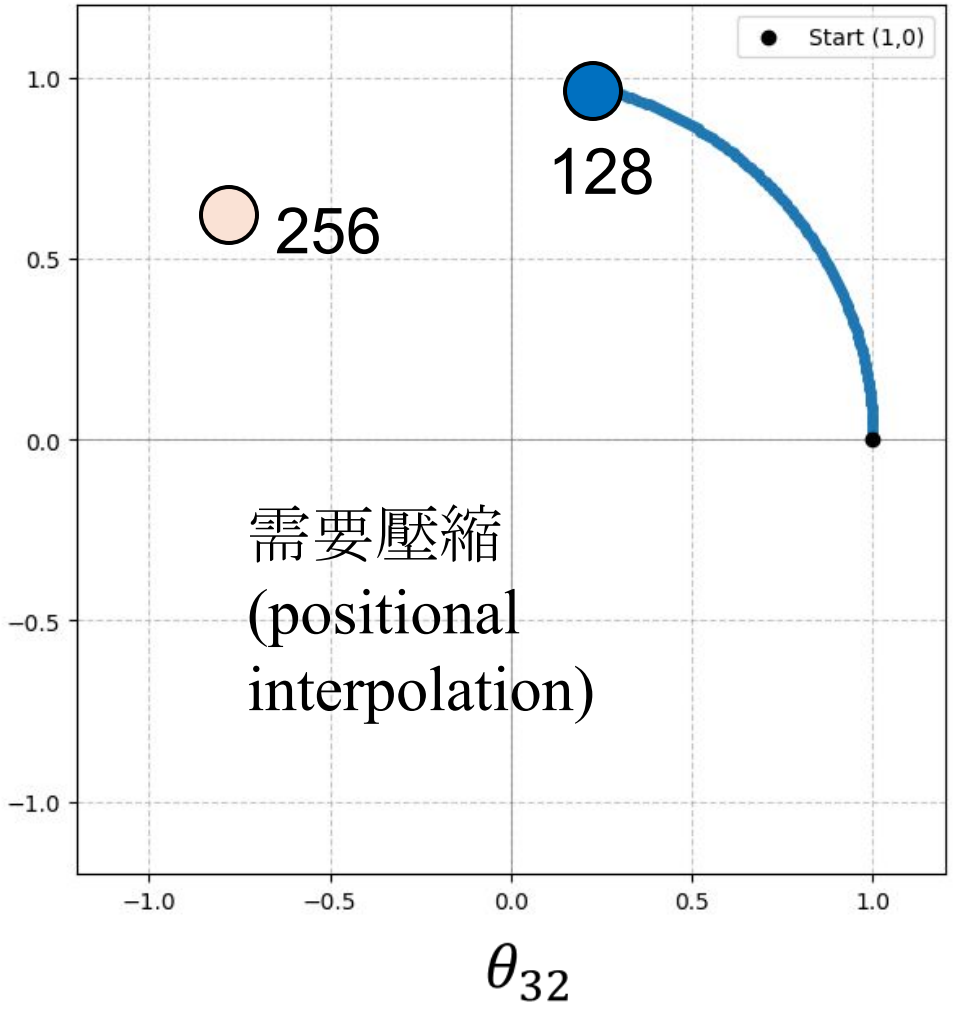
Frequency-Based Approach

$$\theta_i = \frac{1}{10000^{2i/d}}$$

$$i = 0,1,\ldots,\frac{d}{2} - 1$$

$N = 128$



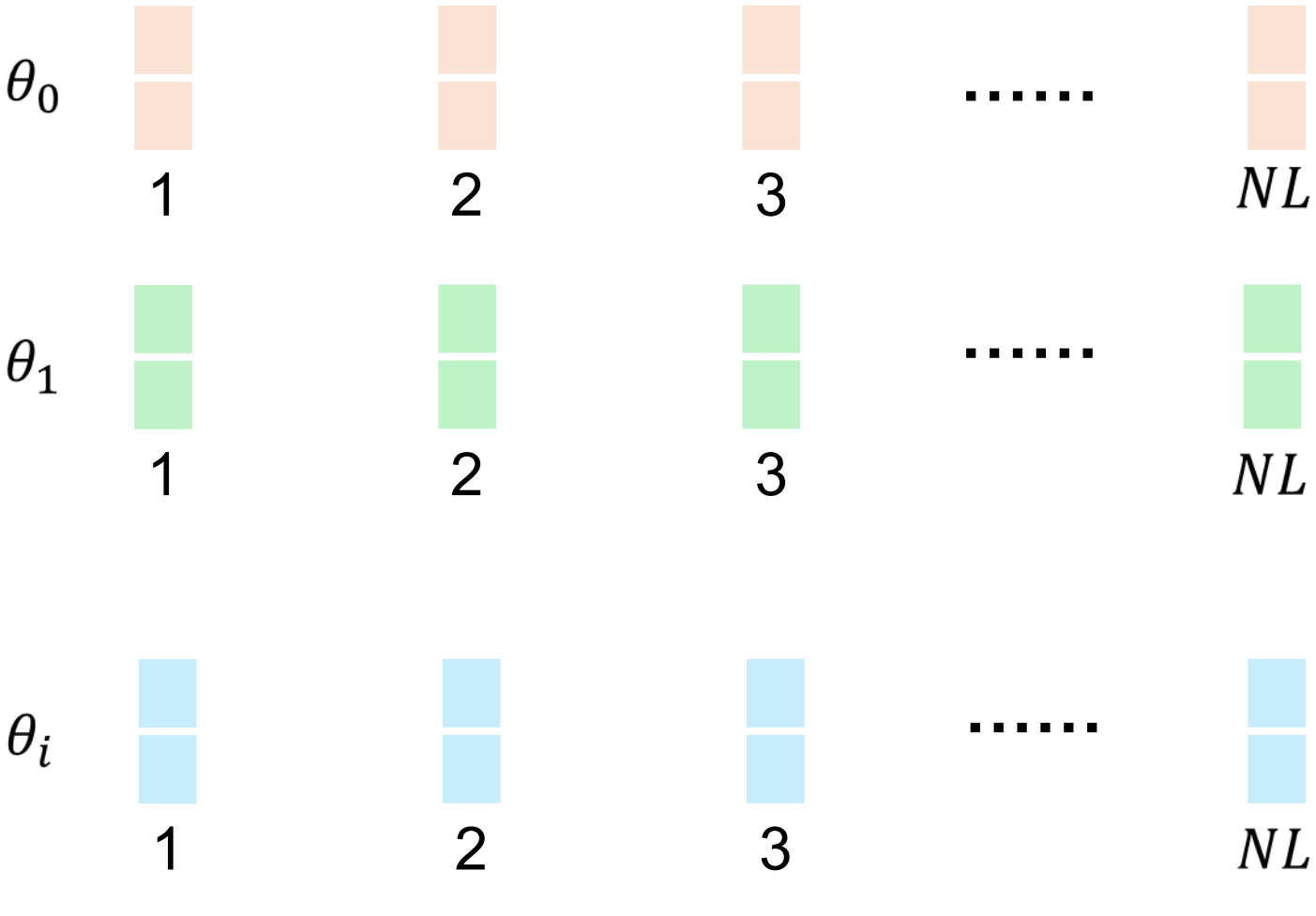


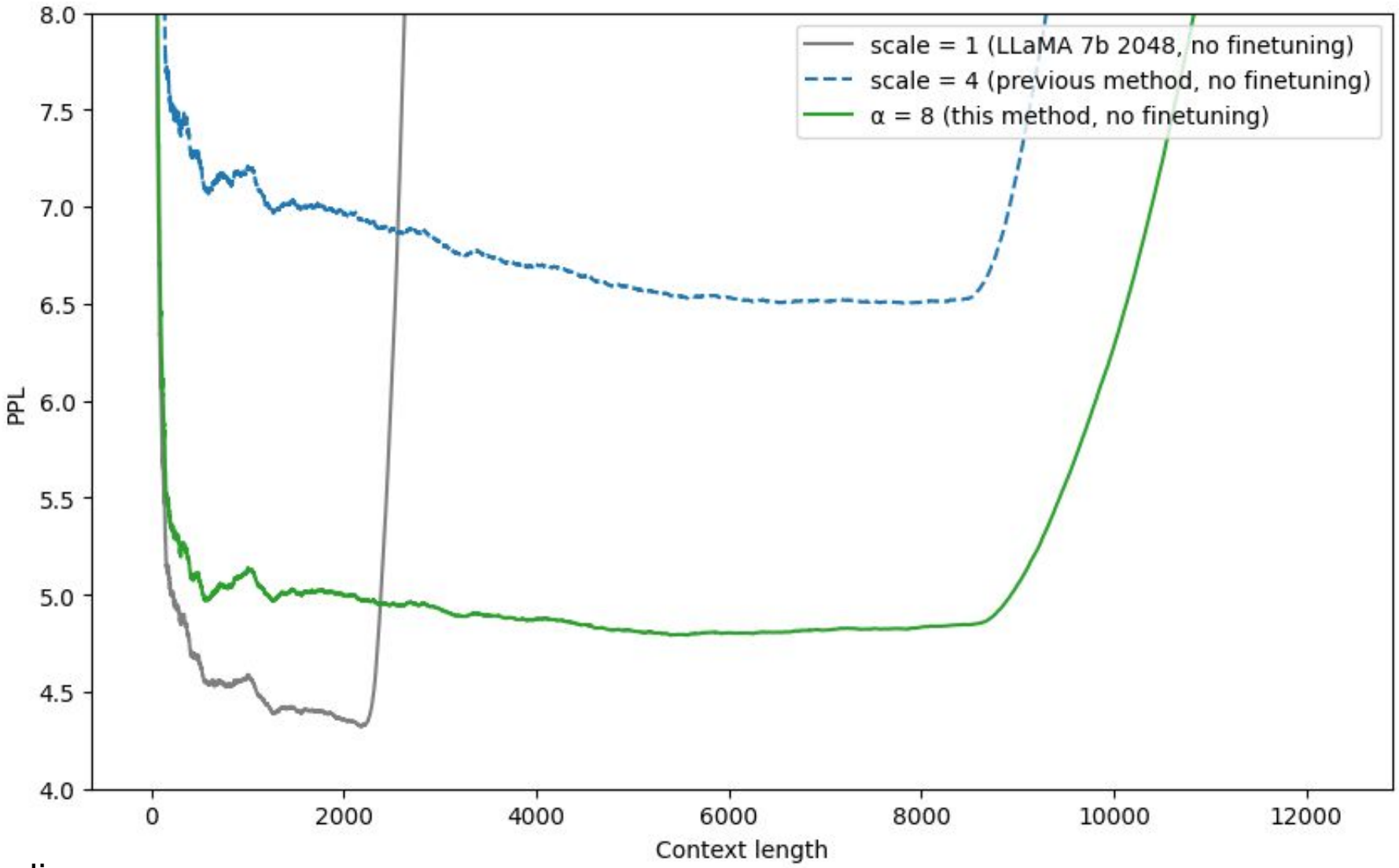
Frequency-Based Approach

NTK-Aware Scaling

$$\theta_i = \frac{1}{10000^{2i/d}}$$

$$i = 0,1,\dots,\frac{d}{2}-1$$

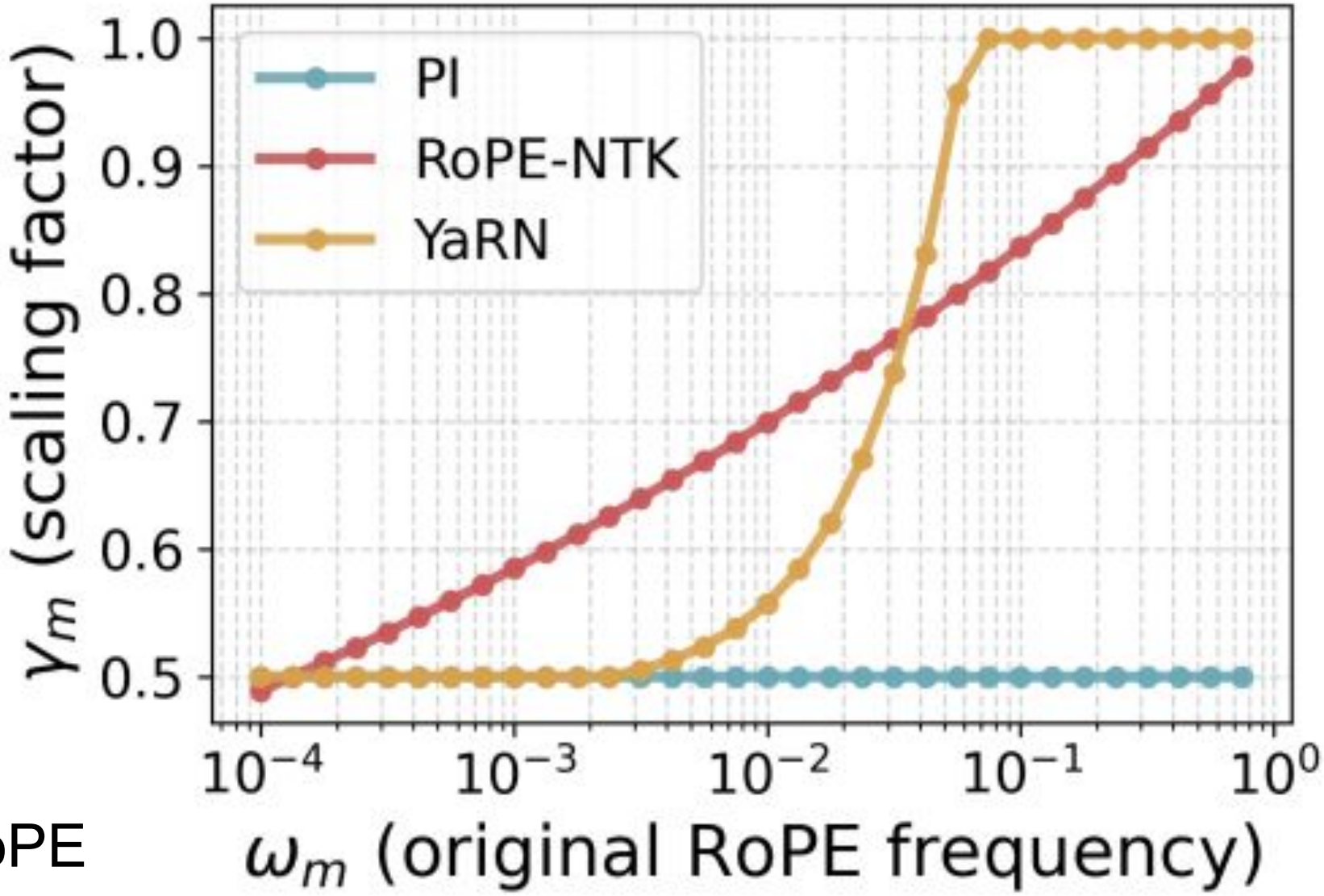




NTK-Aware Scaling

https://www.reddit.com/r/LocalLLaMA/comments/14lz7j5/ntkaware_scaled_rope_allows_llama_models_to_have/

$L = 2$



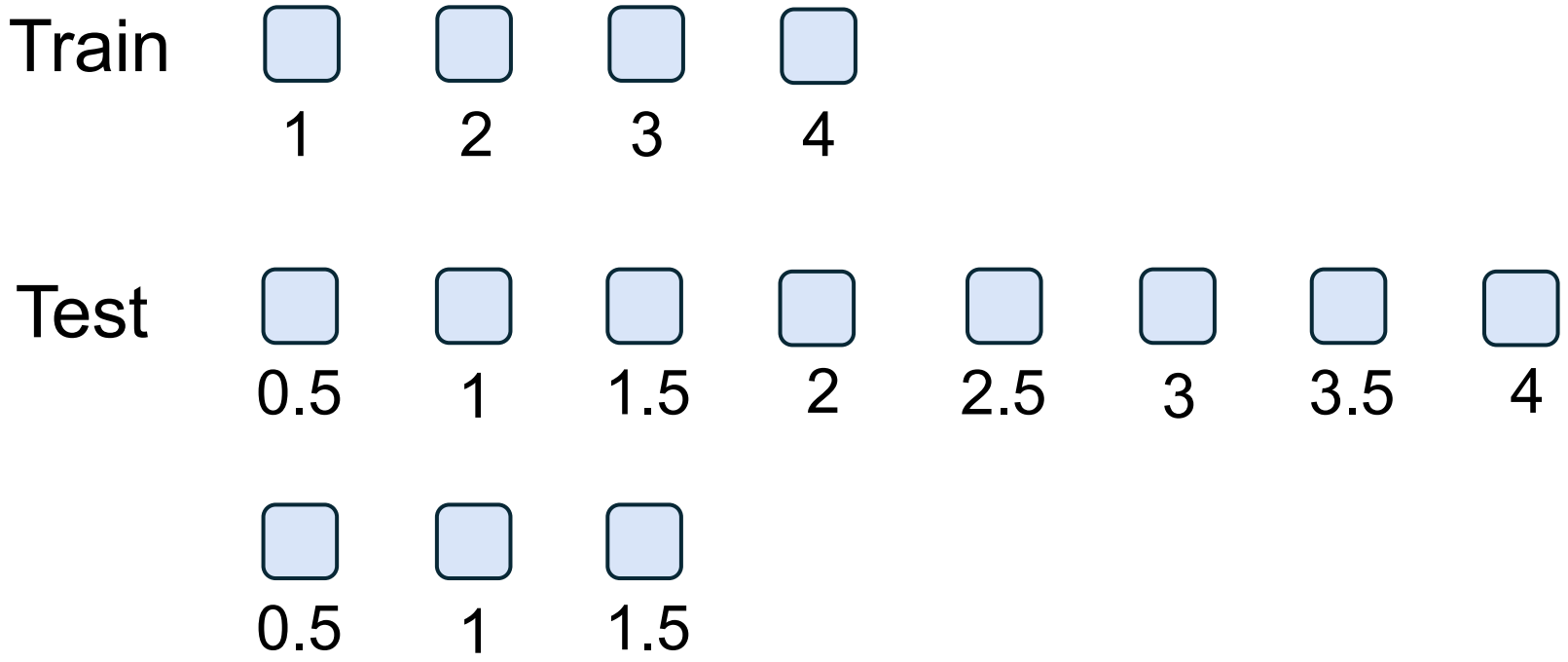
YaRN (Yet another RoPE extensioN method)

https://arxiv.org/abs/2309.00071

Source of figure: https://arxiv.org/pdf/2512.12167

Dynamic Scaling

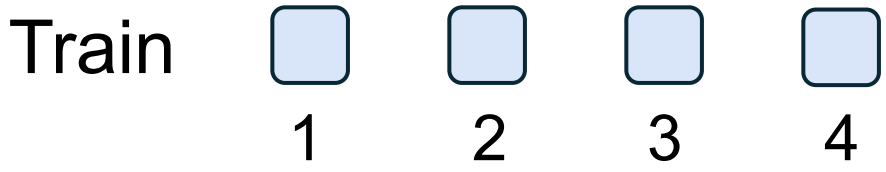
https://www.reddit.com/r/LocalLLaMA/comments/14mrgpr/dynamically_scaled_rope_further_increases/

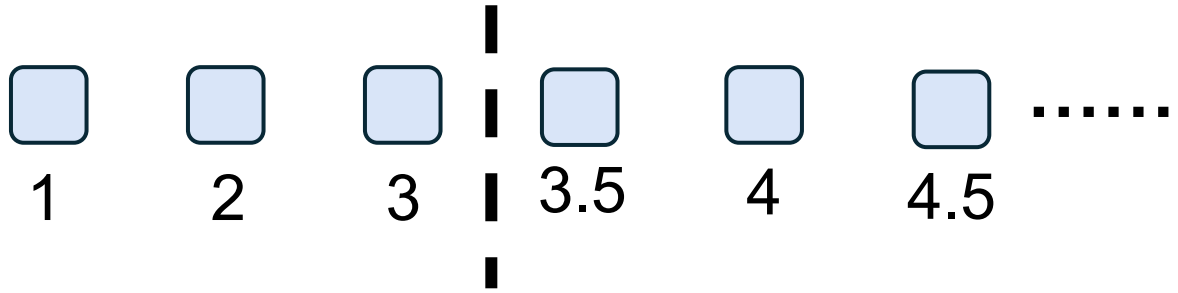


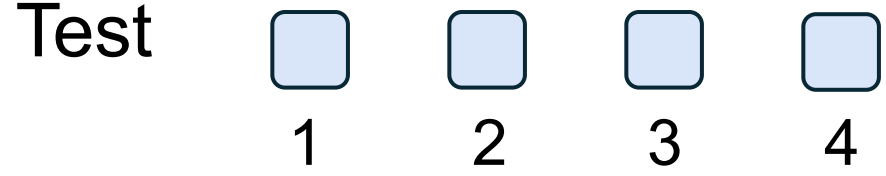
長的序列能做了，但短的序列反而變差了

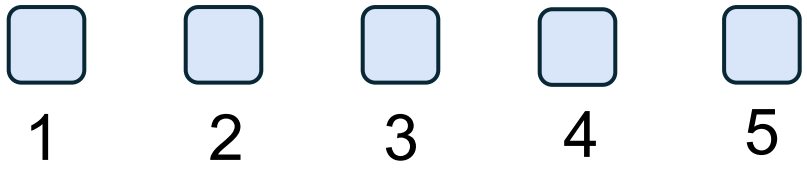
Dynamic Scaling

https://www.reddit.com/r/LocalLLaMA/comments/14mrgpr/dynamically_scaled_rope_further_increases/

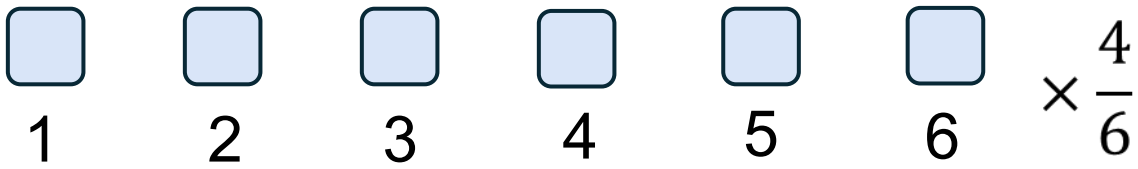






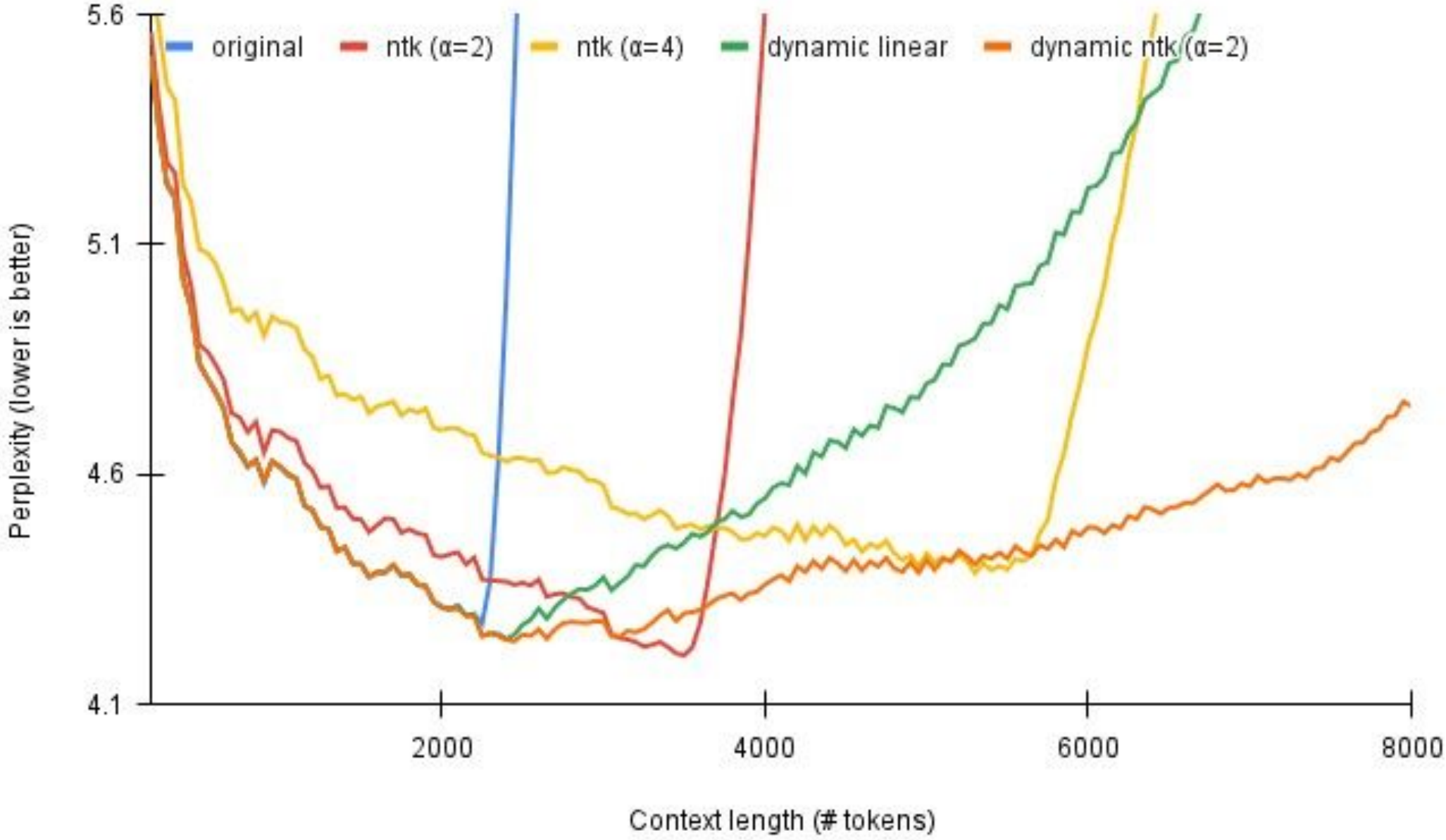


$\times \frac{4}{5}$



$\times \frac{4}{6}$

沒必要壓縮 position 的序列就不壓縮



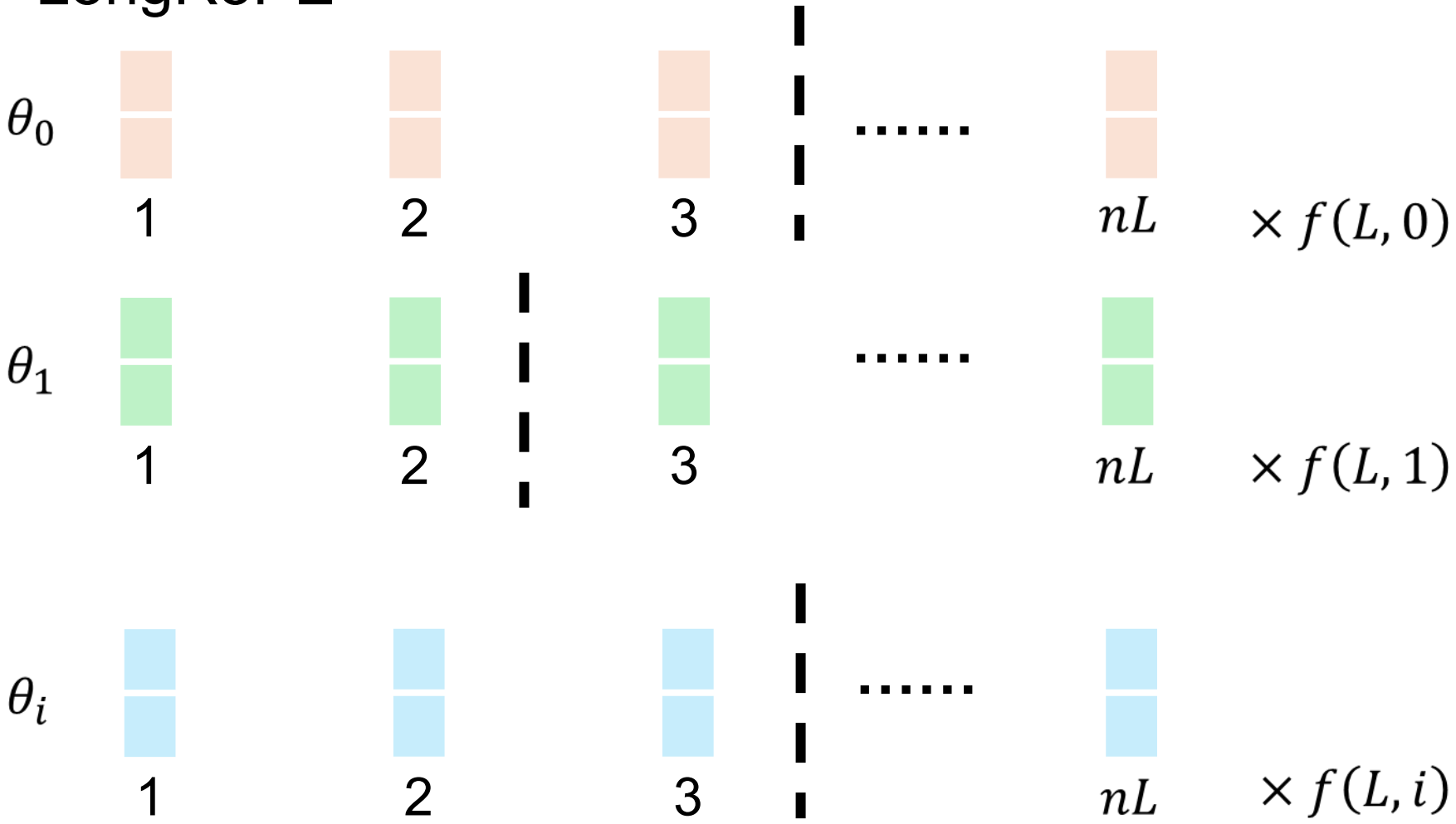
https://www.reddit.com/r/LocalLLaMA/comments/14mrgpr/dynamically_scaled_rope_further_increases/

Frequency-Based + Dynmaic

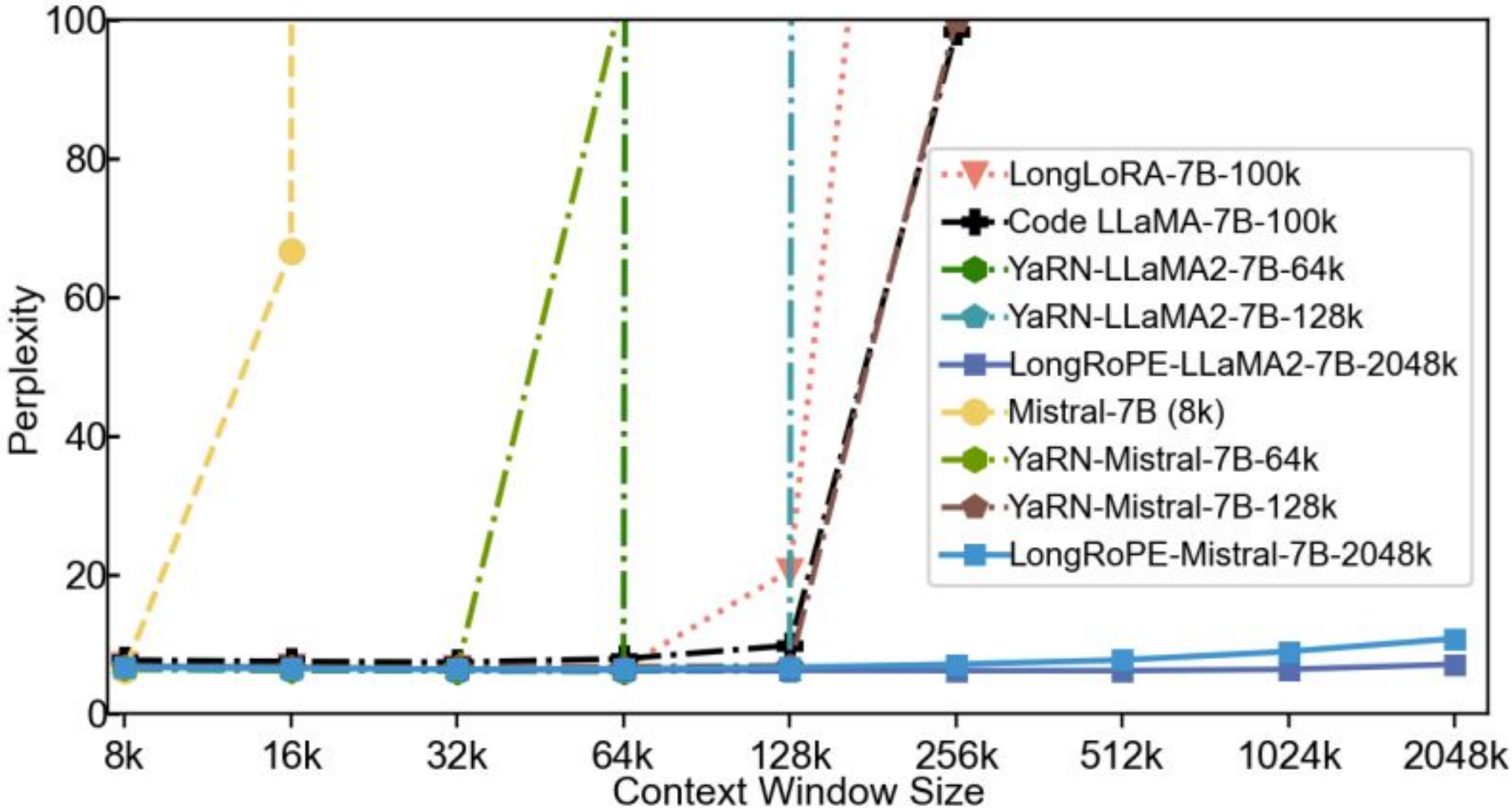
$$\theta_i = \frac{1}{10000^{2i/d}}$$

$$i = 0,1,\ldots,\frac{d}{2} - 1$$

LongRoPE



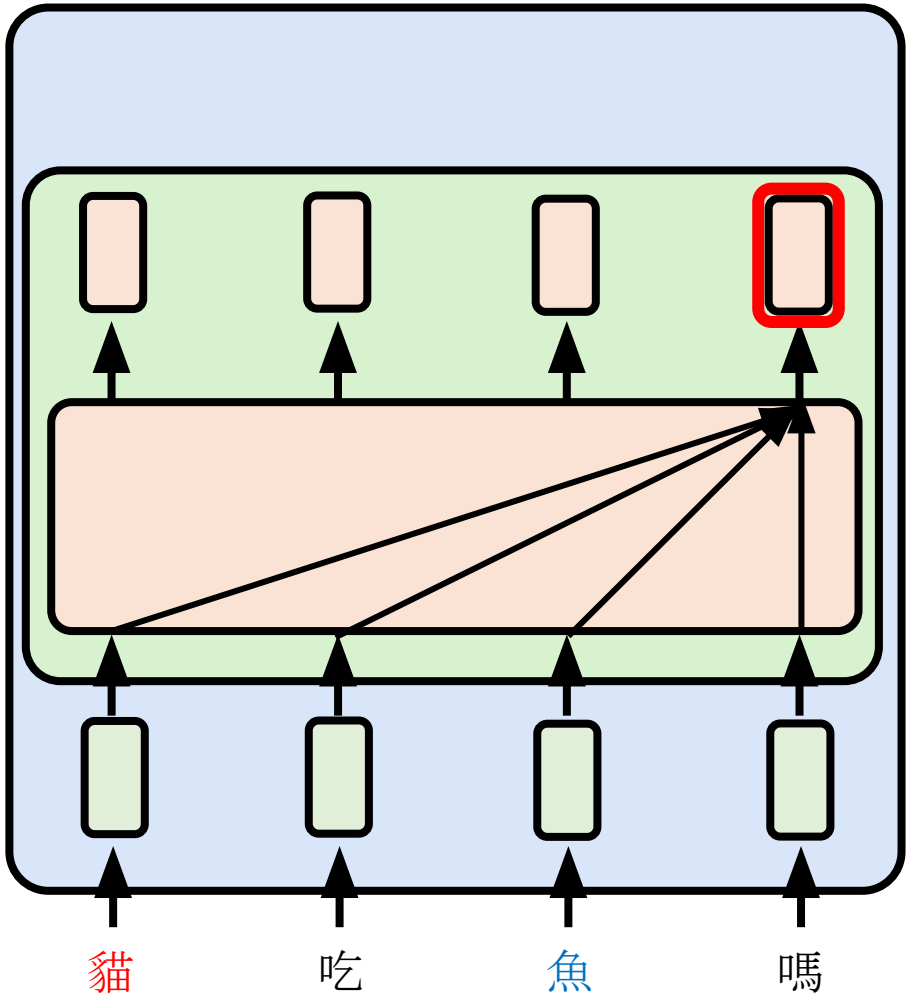
Use evolutionary search

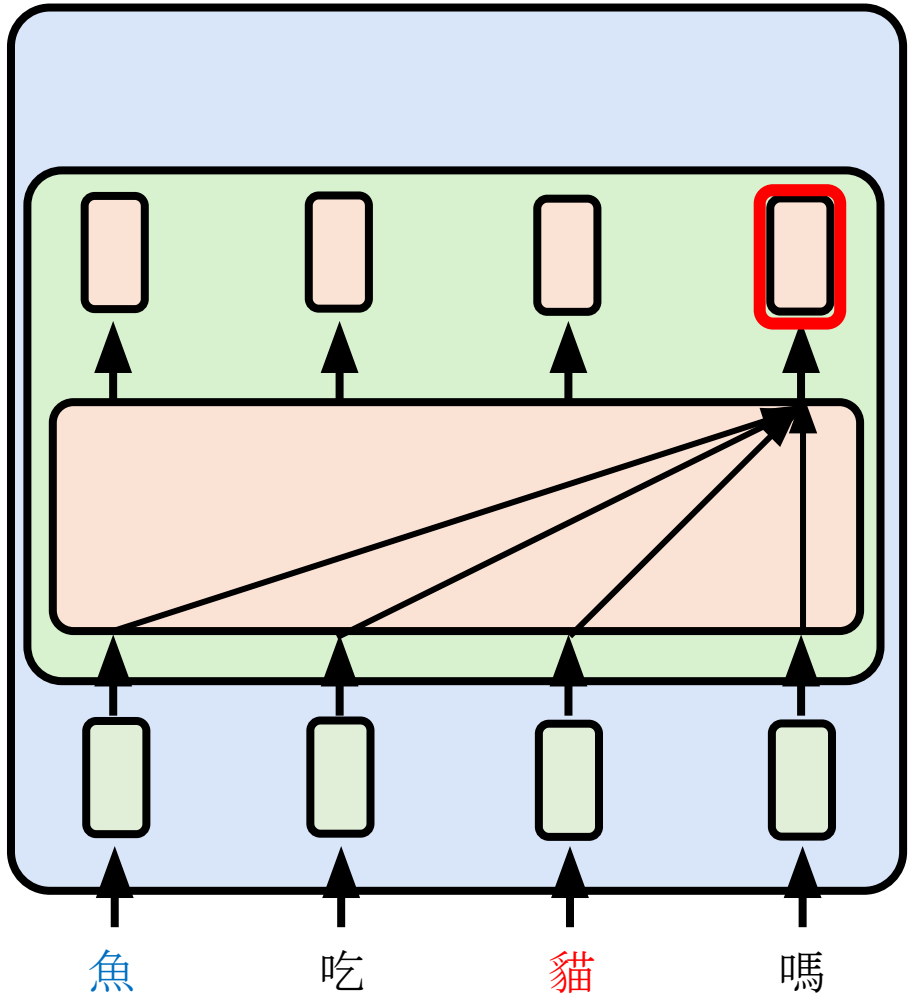


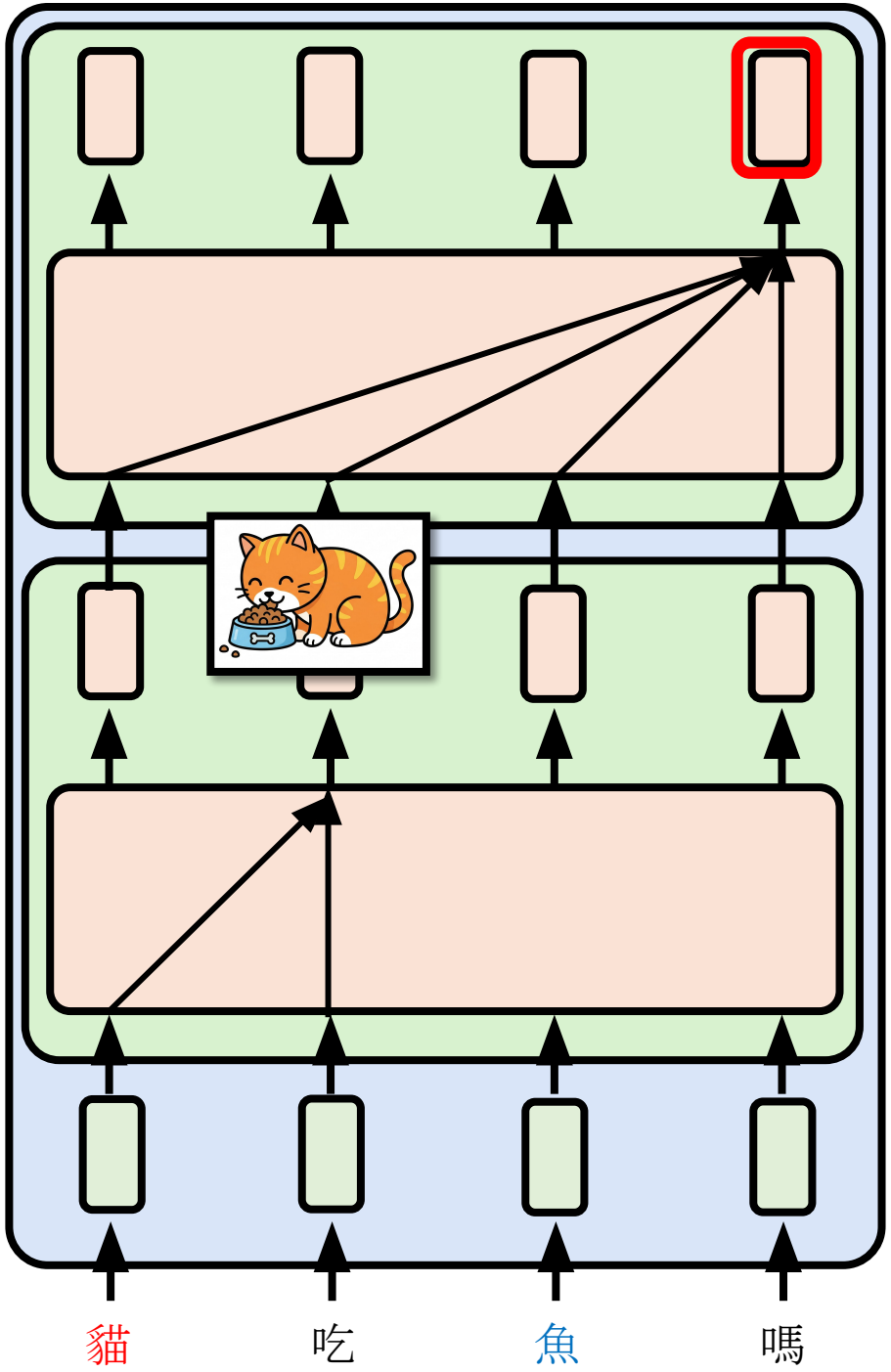
https://arxiv.org/abs/2402.13753

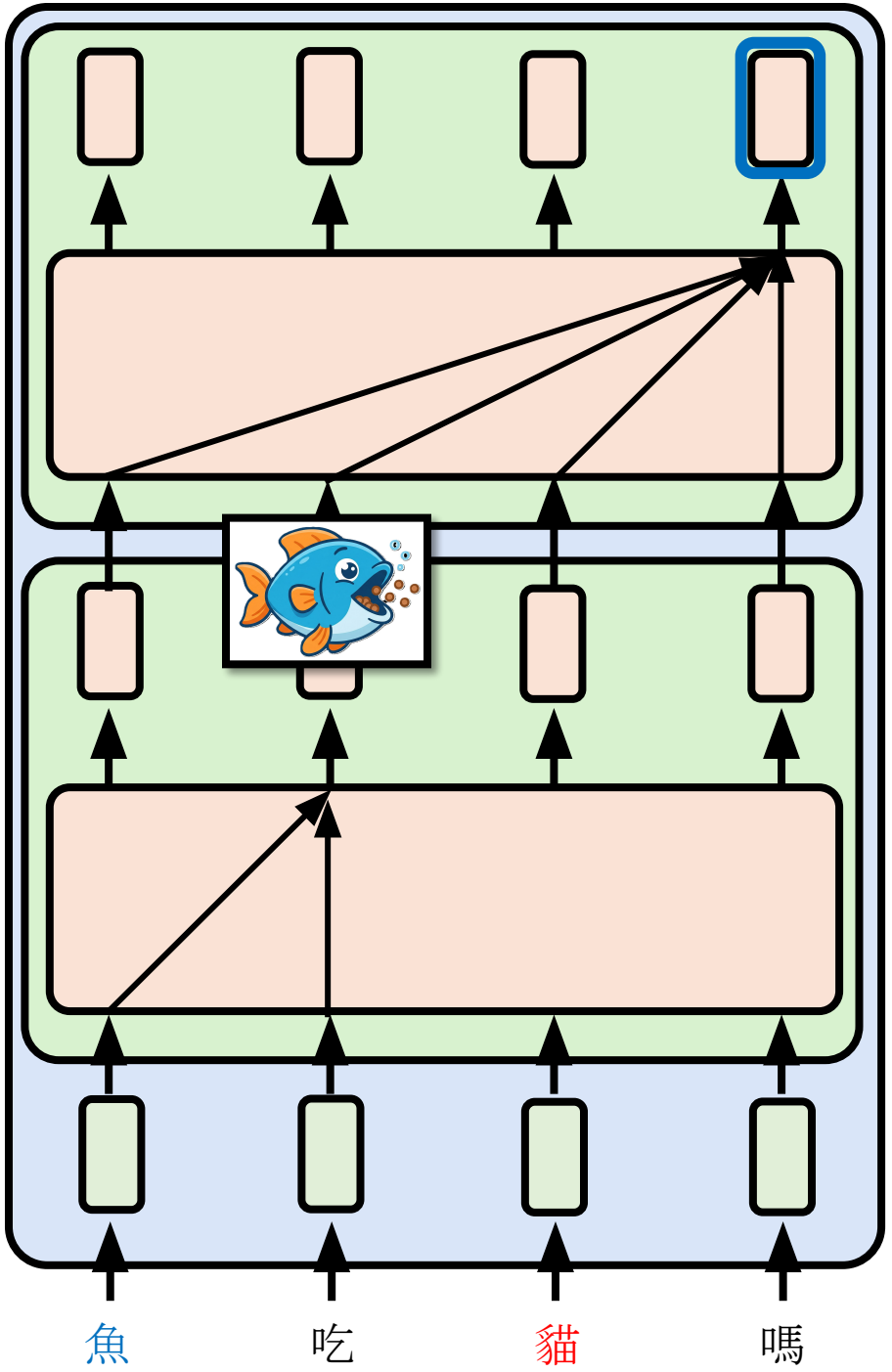
No Positional Embedding?!

Self-attention 真的沒有位置資訊？

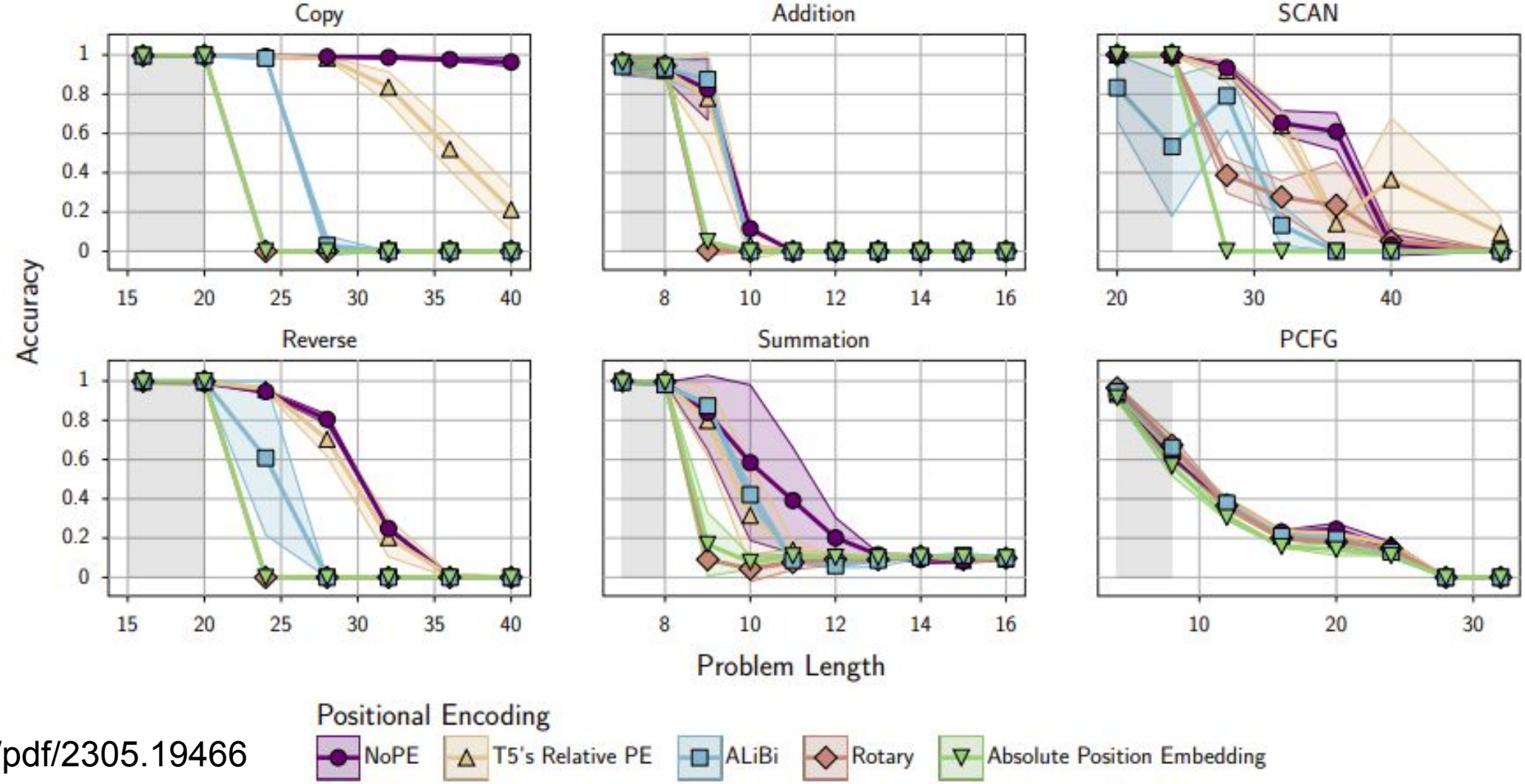





貓
吃
魚
嗎

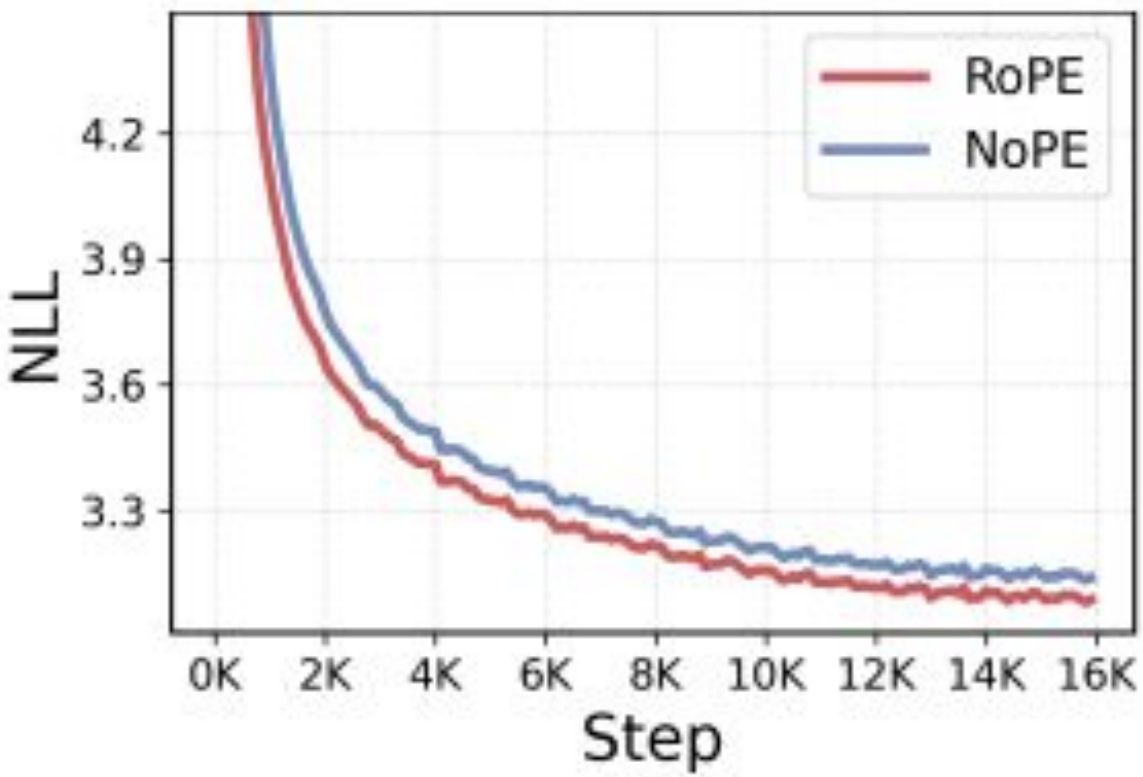

魚
吃
貓
嗎

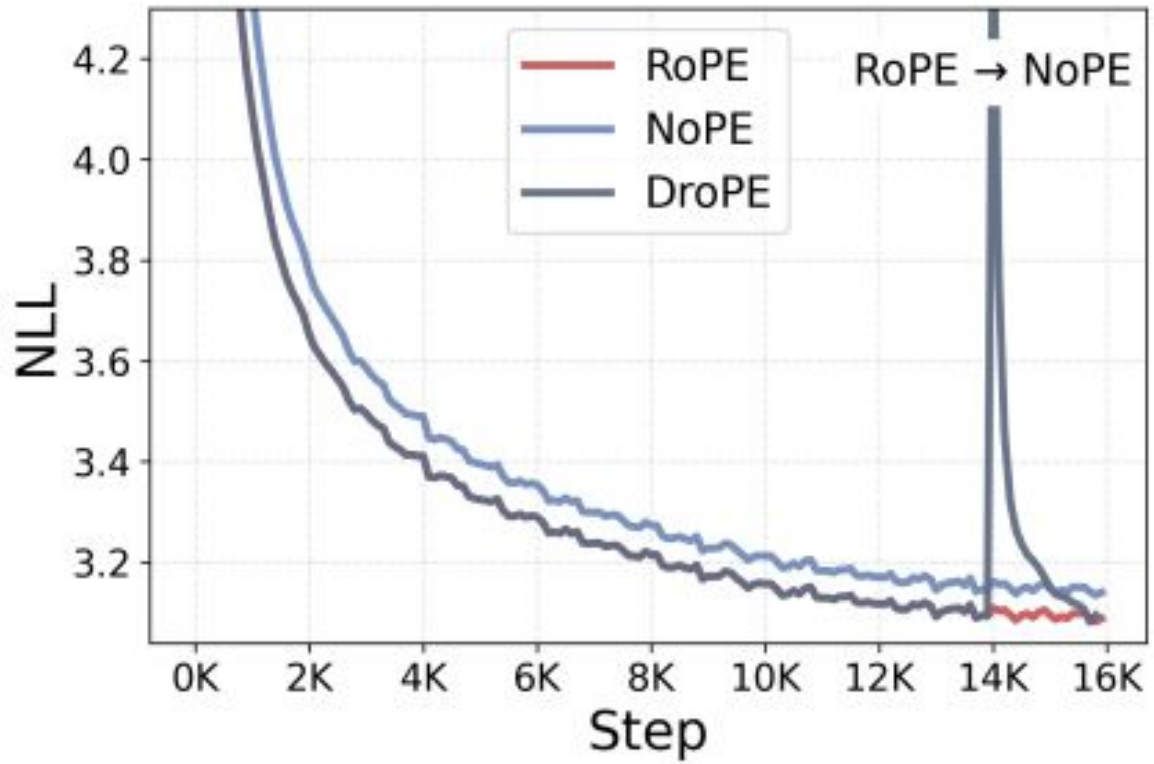
所以沒有必要加 Positional Embedding!



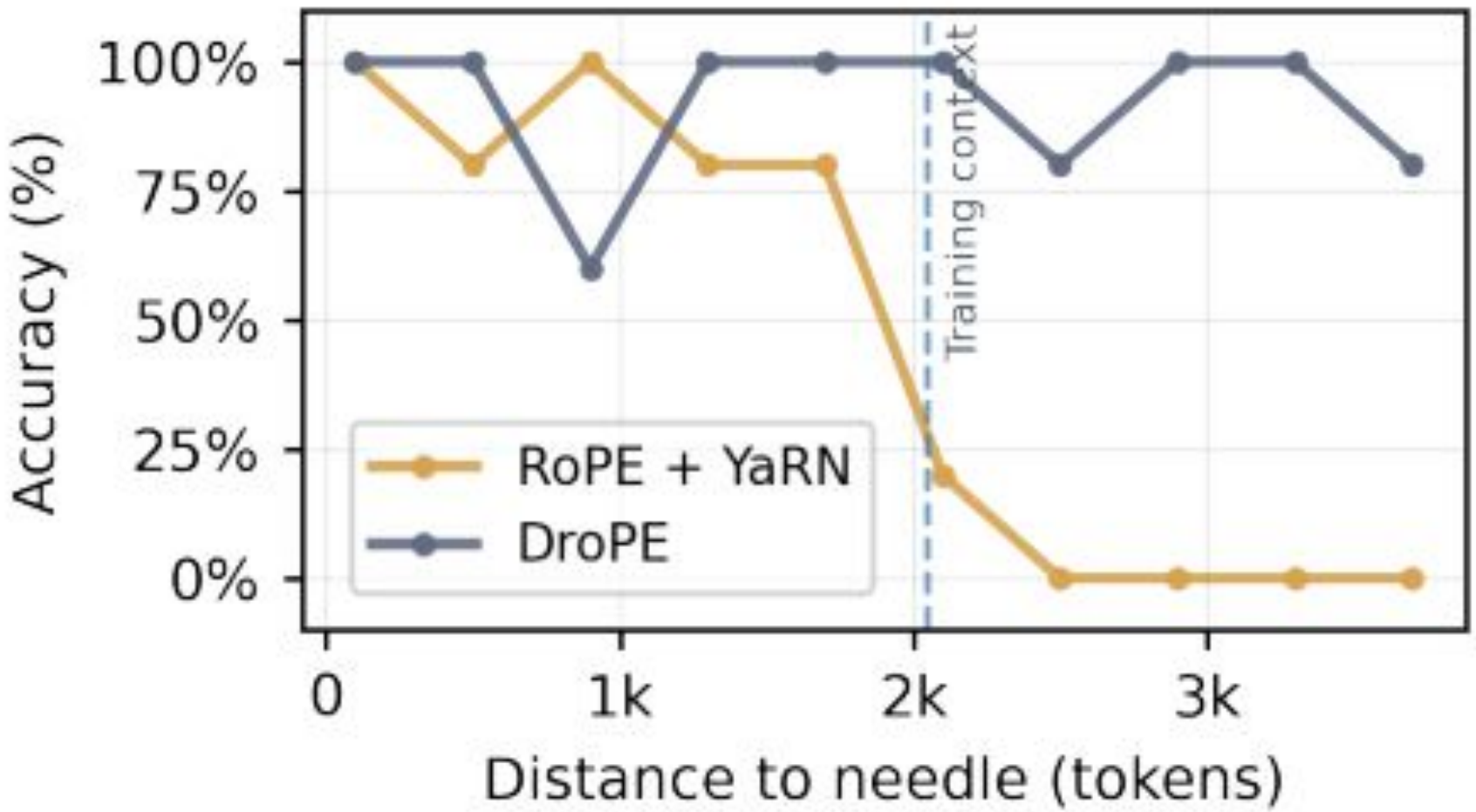
NoPE

https://arxiv.org/pdf/2305.19466





https://arxiv.org/pdf/2512.12167



https://arxiv.org/pdf/2512.12167

Concluding Remarks

- Absolute Positional Embedding
- Relative Positional Embedding
- RoPE
- Train short, test long
- No Positional Embedding?!